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A 3.6mW 5MS/s 100dB-SNDR 2-Step SAR ADC with Continuous-Time Second-Stage and Tracking Averaging

Congyu Jin, Zhaoyang Shen, Xingshuai Zou, Xuanhao Zhang, Han Zhao, Zhong Zhang, Ning Ning, Hongshuai Zhang, Zheng Wang, Zhenbing Li, Shiheng Yang, Xiangyu Mao, Jiaxin Liu

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Outline

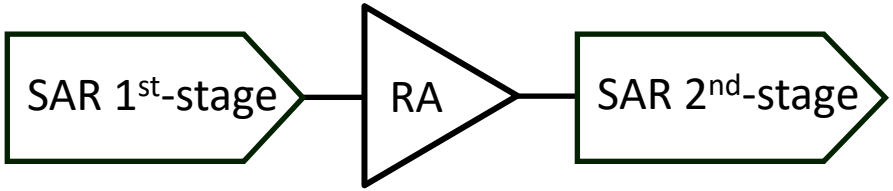
- Motivation
- Proposed 2-Step SAR with CT 2nd-Stage
- Circuit Implementation
- Measurement Results
- Conclusion

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- **Motivation**
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2-Step SAR ADC

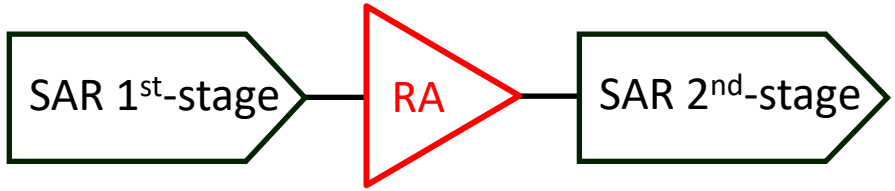
- High speed (＞1MSPs)
- High resolution (＞16bit)
- Low latency
- Excellent power efficiency



- **Design Challenge: Residue Amplifier**
  - High speed
  - Large output swing
  - Good linearity
  - Low noise

2-Step SAR ADC

- High speed (＞1MSPs)
- High resolution (＞16bit)
- Low latency
- Excellent power efficiency

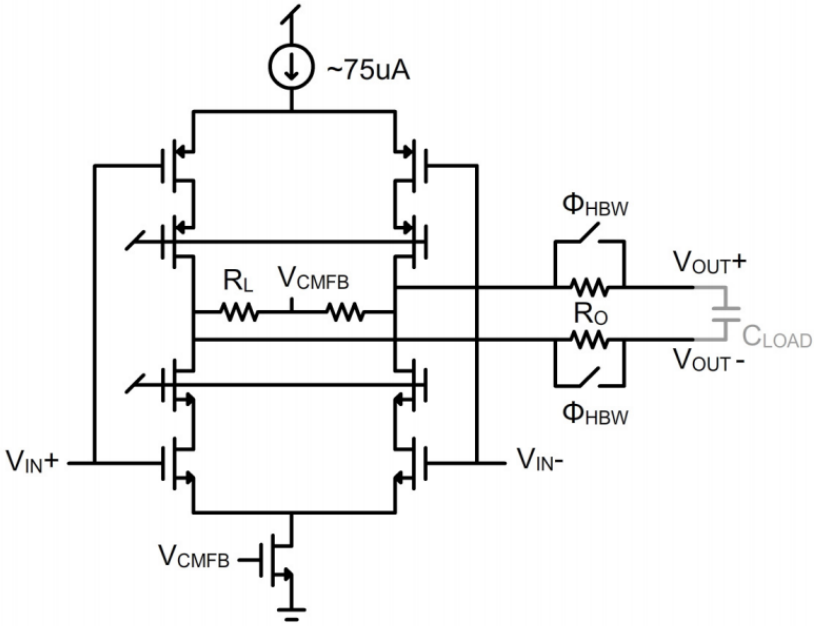


- **Design Challenge: Residue Amplifier**
  - High speed
  - Large output swing
  - Good linearity
  - Low noise

Low-noise, high-speed RA is power-hungry

Prior Solution to Reduce RA Noise

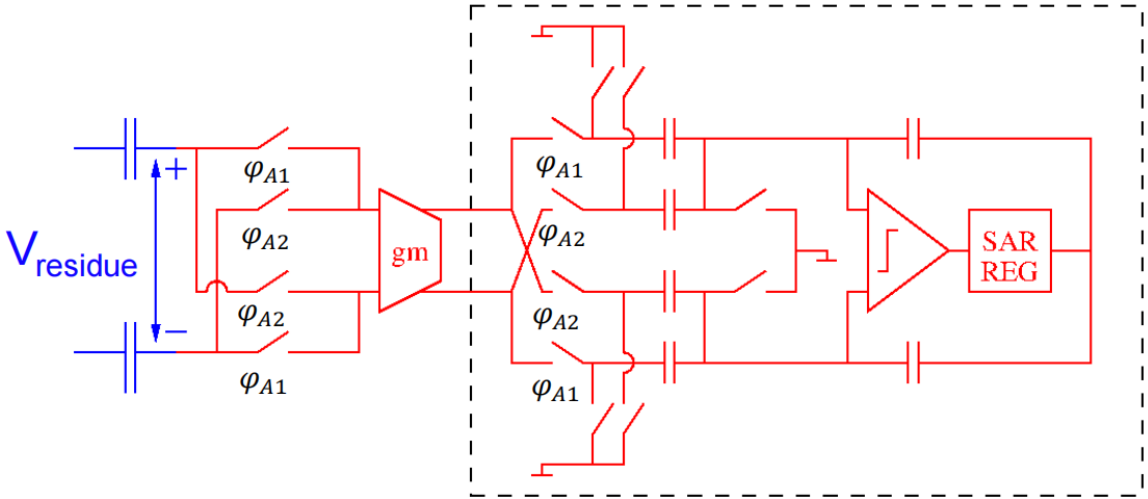
- Two Phase Settling
  - Limited noise reduction performance (~34%)
  - May affect RA settling

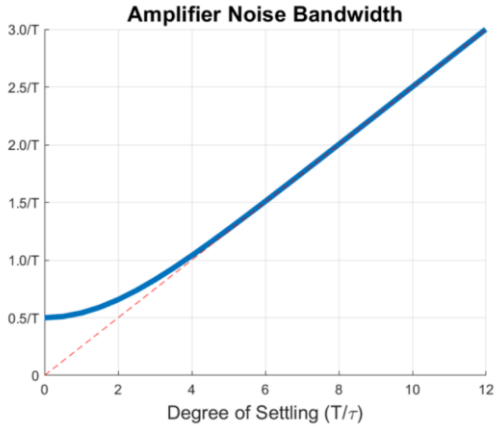


[ISSCC 2020, L. Jie]

Prior Solution to Reduce RA Noise

- Integrating RA
  - PVT sensitive voltage gain
  - Need gain calibration

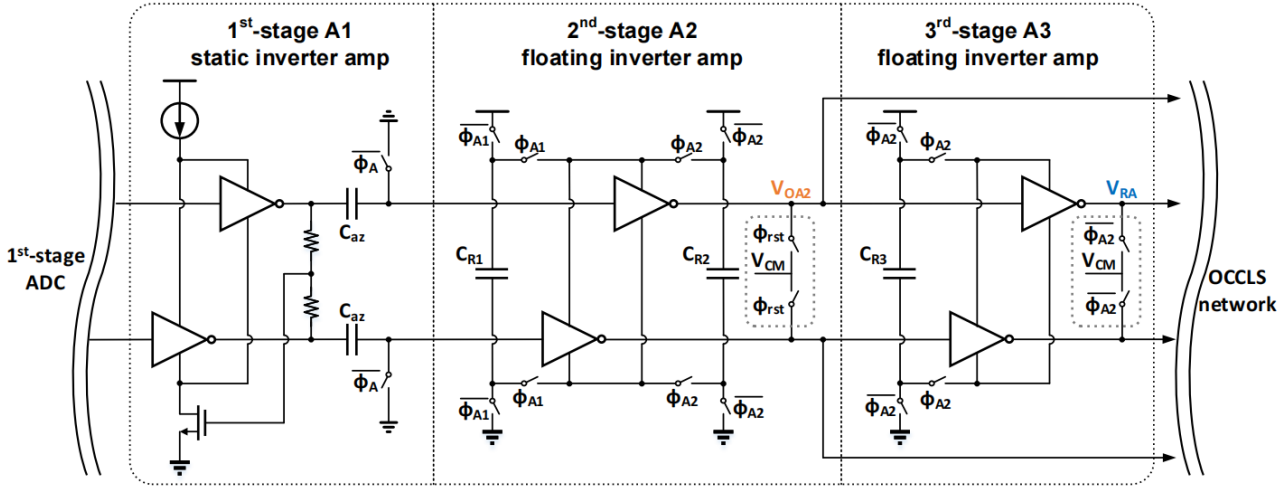


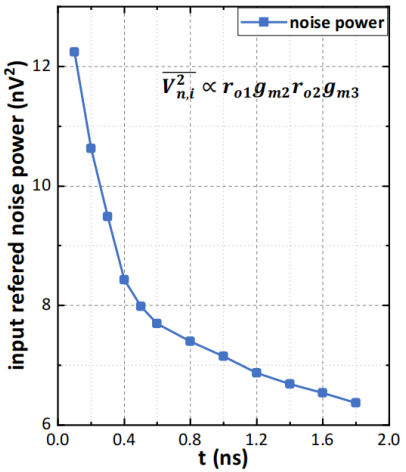


[ISSCC 2022, J. Steensgaard]

Prior Solution to Reduce RA Noise

- Bandwidth adaptive RA
  - Limited output swing





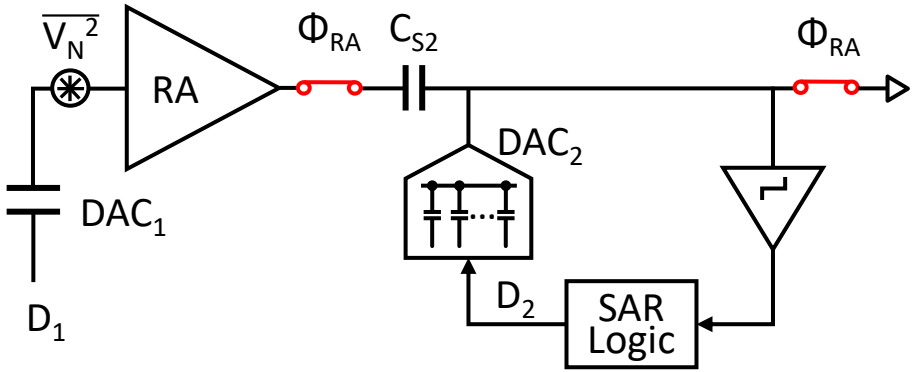
[ISSCC 2024, S. Ye]

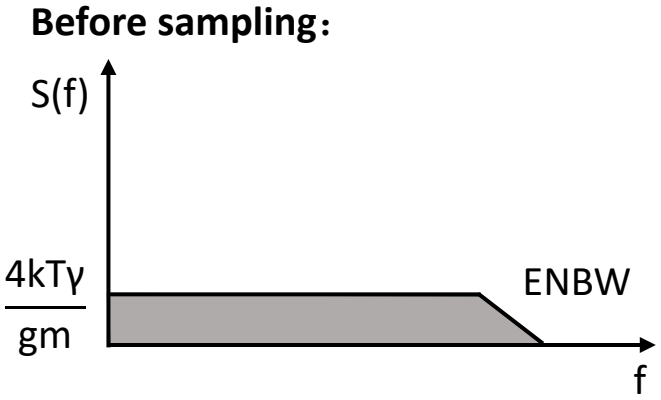
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- **Proposed 2-Step SAR with CT 2nd-Stage**
- Circuit Implementation
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Traditional 2-Step SAR with DT 2nd-Stage

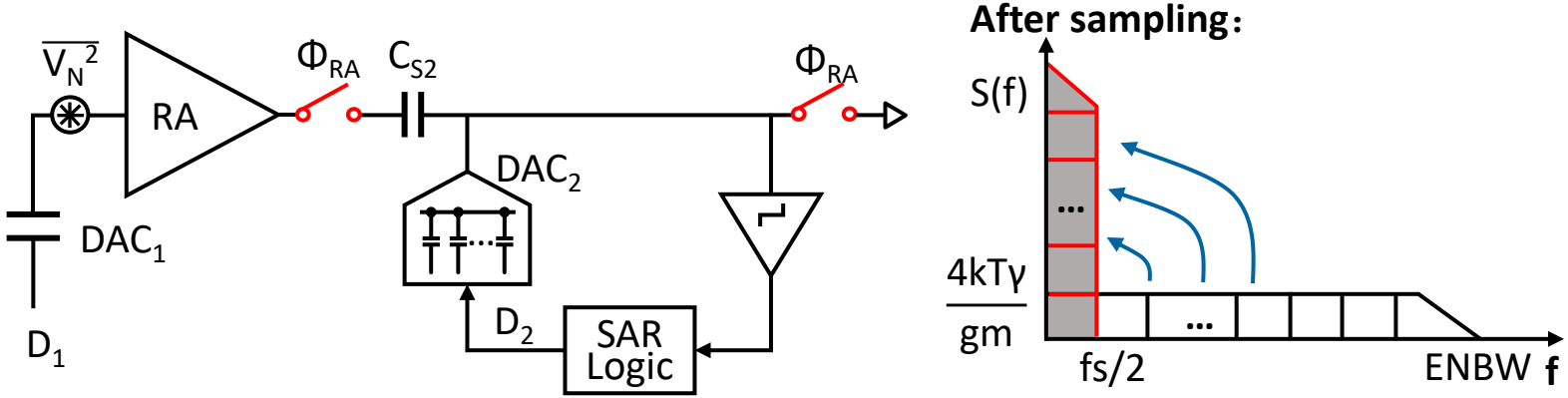
- RA noise sampled on C_{S2}
- $\overline{V_{n,RA}^2} \approx \frac{4kT\gamma}{g_m} \cdot ENBW$





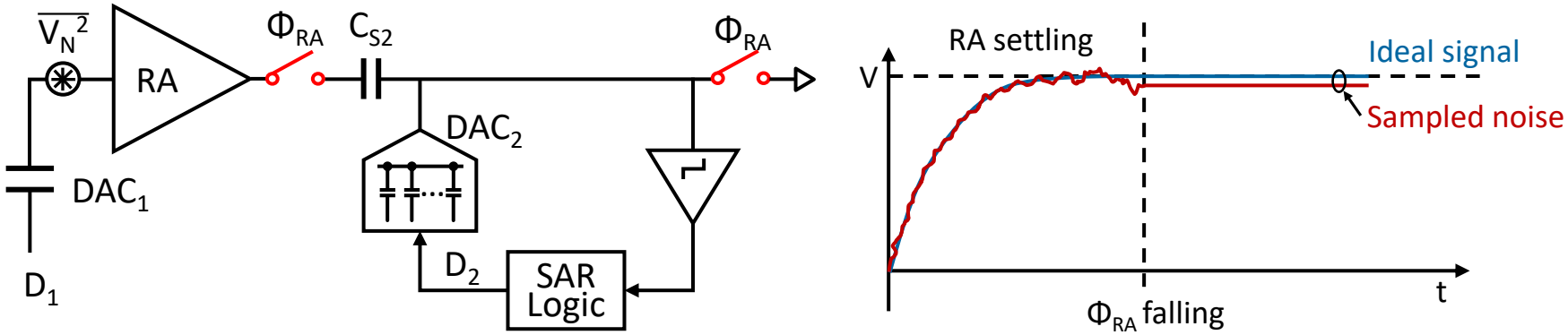
Traditional 2-Step SAR with DT 2nd-Stage

- RA noise aliased to in-band



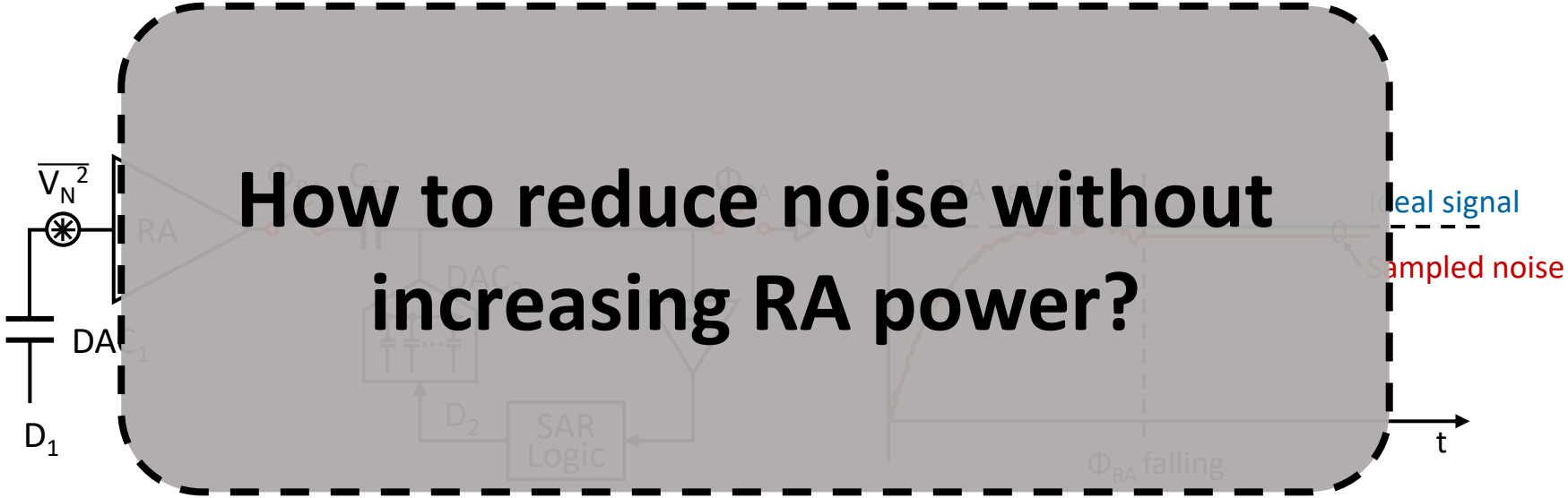
Traditional 2-Step SAR with DT 2nd-Stage

- Sampled RA noise limits the overall ADC SNR



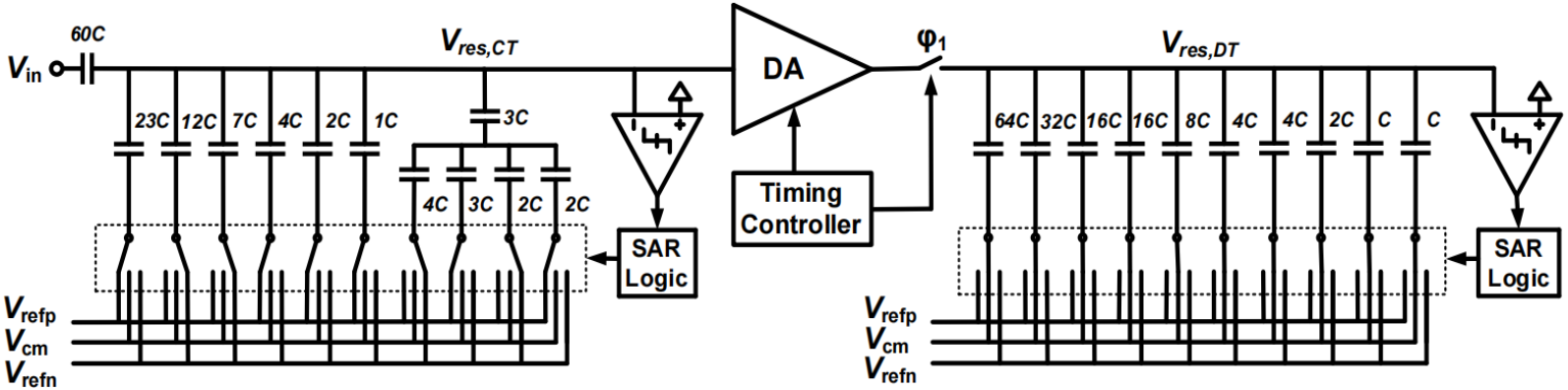
Traditional 2-Step SAR with DT 2nd-Stage

- Sampled RA noise limits the overall ADC SNR



Prior Works: CT SAR for 1st-Stage

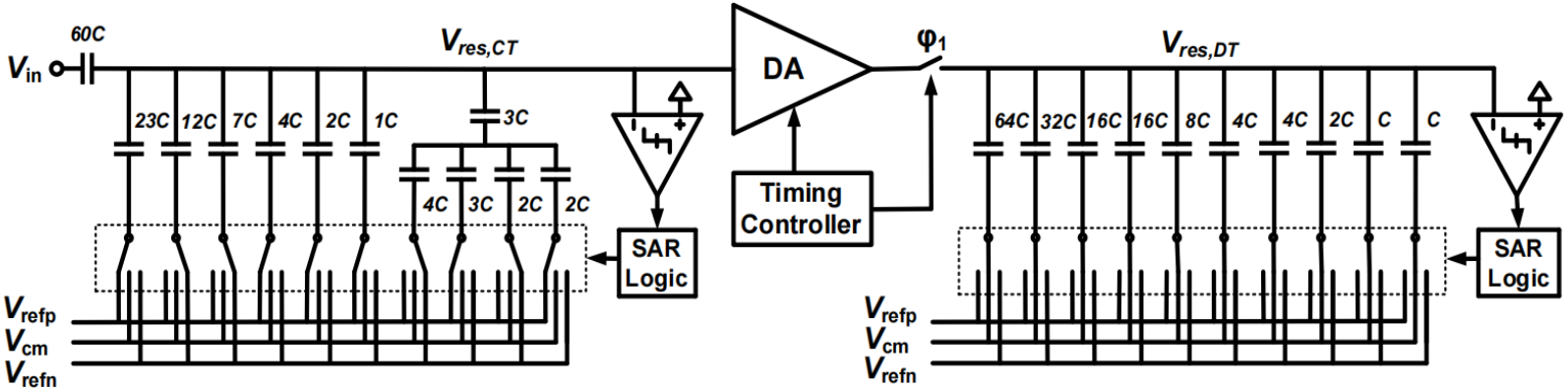
- V_{IN} couples to CDAC without sampling switch
- Open-loop RA transfers V_{RES1} to the DT 2nd-stage



[ISSCC 2019, L. Shen]

Prior Works: CT SAR for 1st-Stage

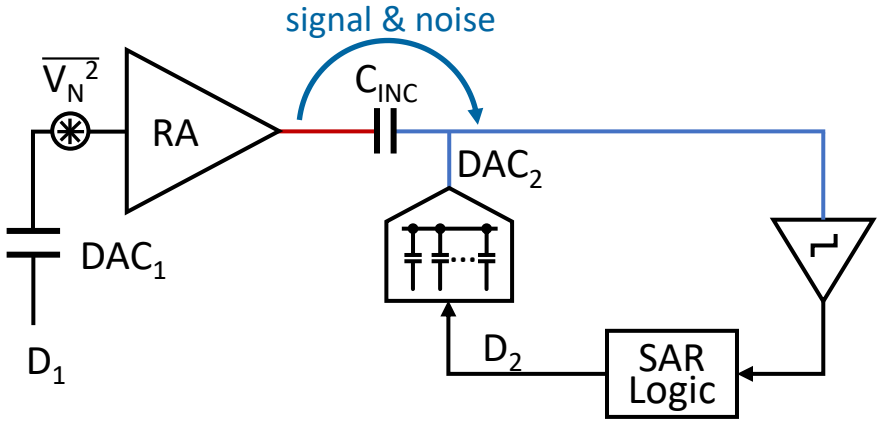
- No sampling noise in 1st-stage
- Suffer from V_{IN} variation ➡ Need high speed SAR logic and RA
  (At 2 MS/s, only 500ps/cycle)

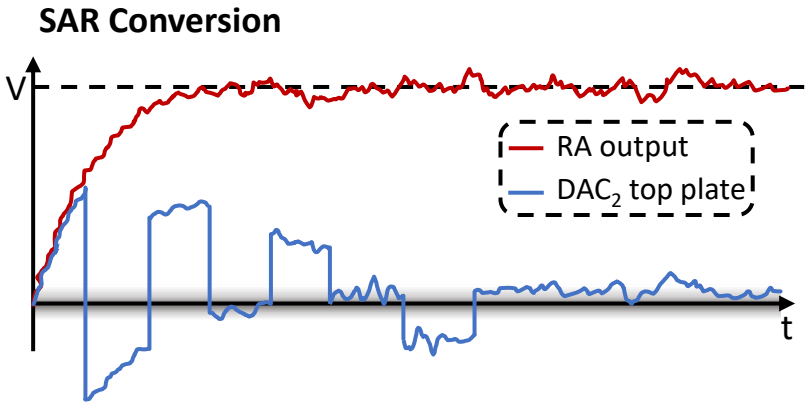


[ISSCC 2019, L. Shen]

Our Solution: CT SAR for 2nd-Stage

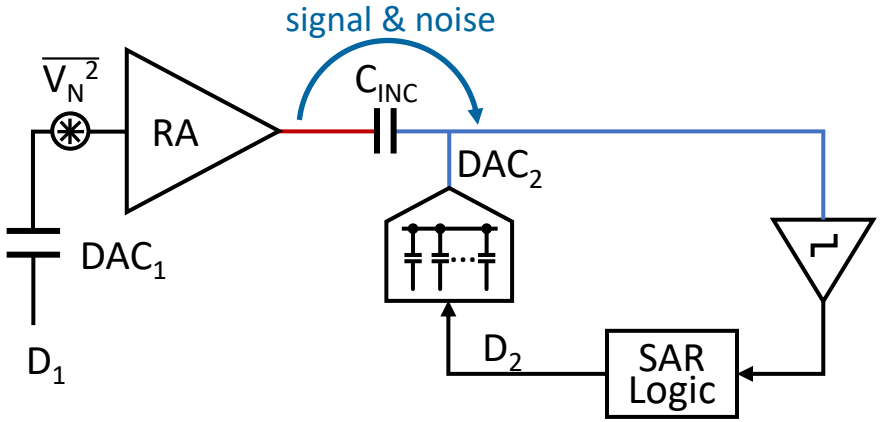
- 2nd-stage sampling switch removed
- CT SAR quantizes amplified V_{RES1} ➡ Quasi-static RA output

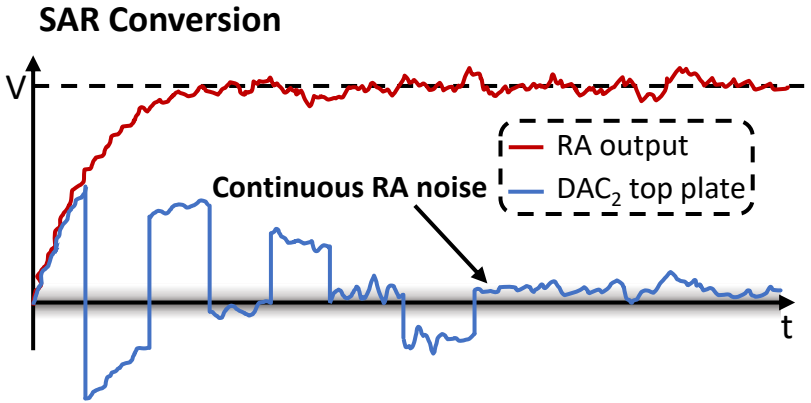




Our Solution: CT SAR for 2nd-Stage

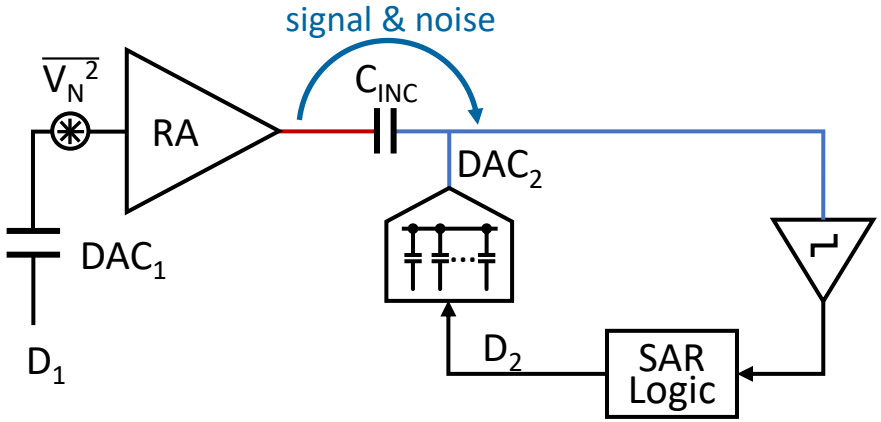
- RA noise continuously couples to DAC_2
- No in-band RA noise aliasing
- RA noise averaging becomes possible

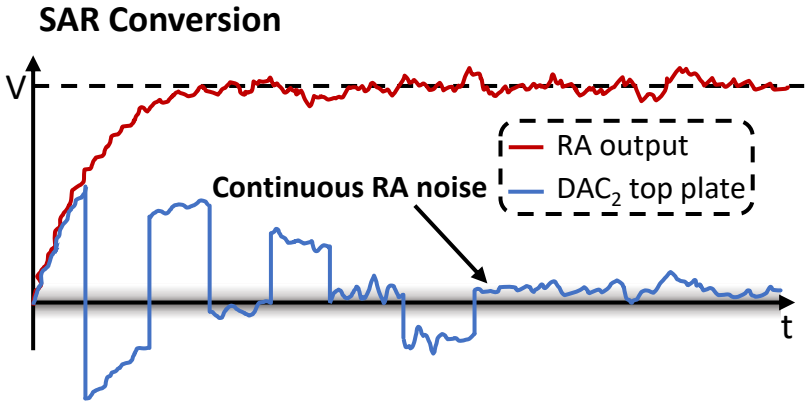




Our Solution: CT SAR for 2nd-Stage

- RA noise continuously couples to DAC_2
- No in-band RA noise aliasing
- RA noise averaging becomes possible **How?**

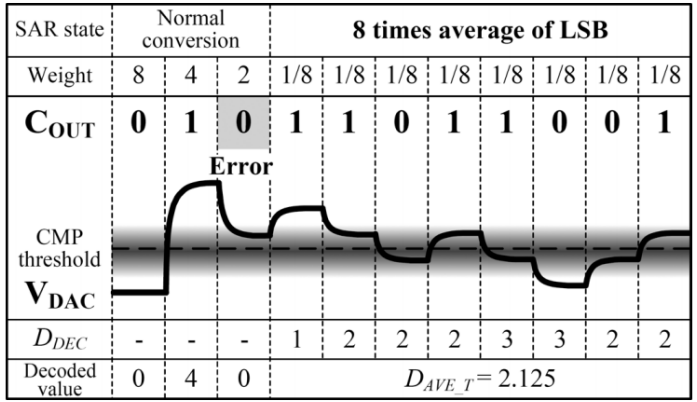




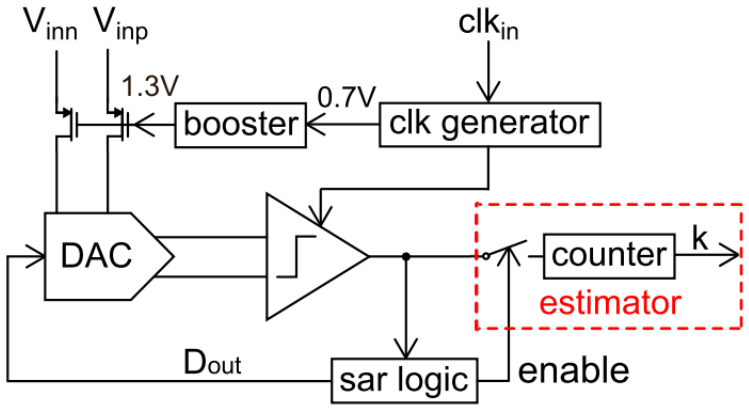
Prior Works: LSB Repeating

- Reducing comparator and reference noise through repeating

SAR state	Normal conversion			8 times average of LSB							
Weight	8	4	2	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8
C_{OUT}	0	1	0	1	1	0	1	1	0	0	1
D_{DEC}	-	-	-	1	2	2	2	3	3	2	2
Decoded value	0	4	0	D_{AVE_T} = 2.125							



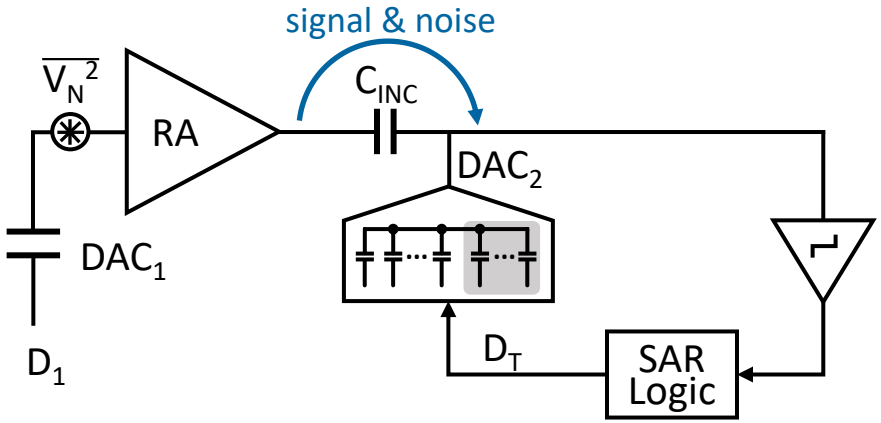
[JSSC 2015, T. Miki]

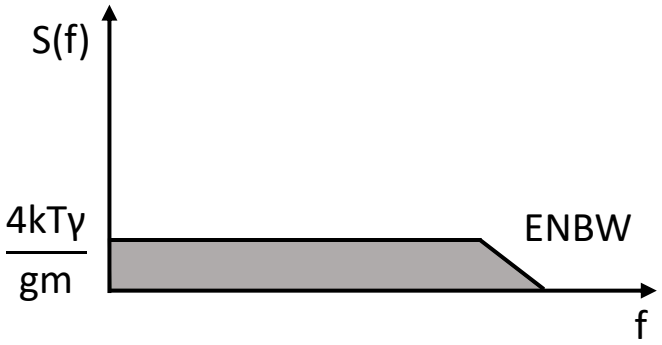


[JSSC 2017, L. Chen]

Proposed CT 2nd-Stage with LSB repeating

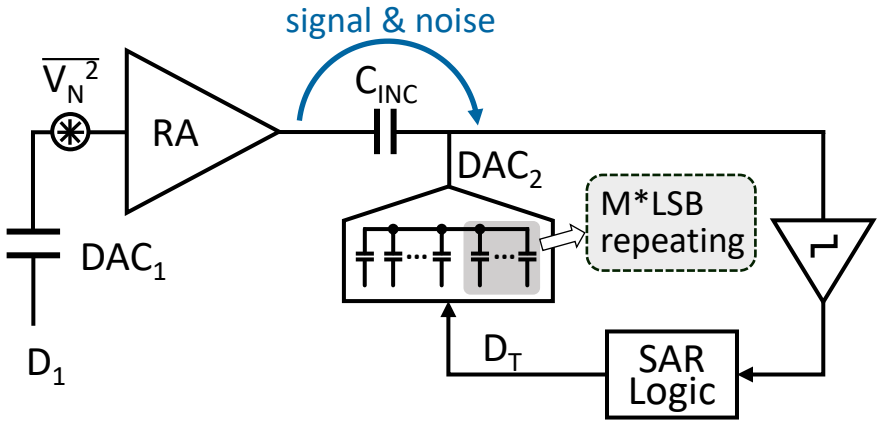
- RA noise is equivalent to comparator noise in CT 2nd-stage

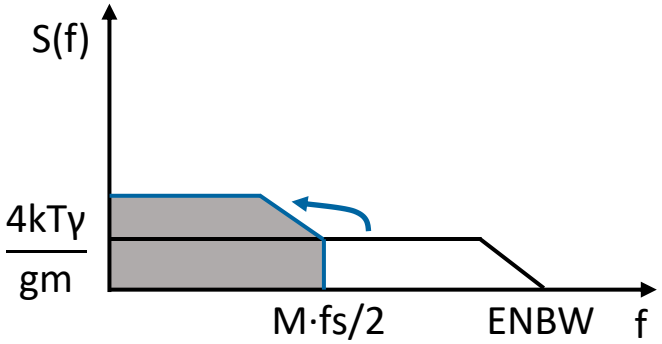




Proposed CT 2nd-Stage with LSB repeating

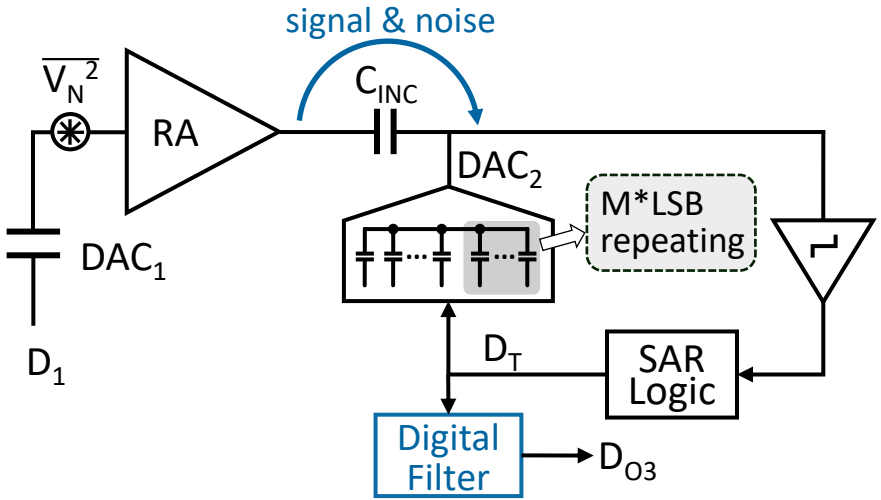
- DAC_2 performs M times LSB repeating
- LSB repeating shifts noise-folding band to M·fs/2

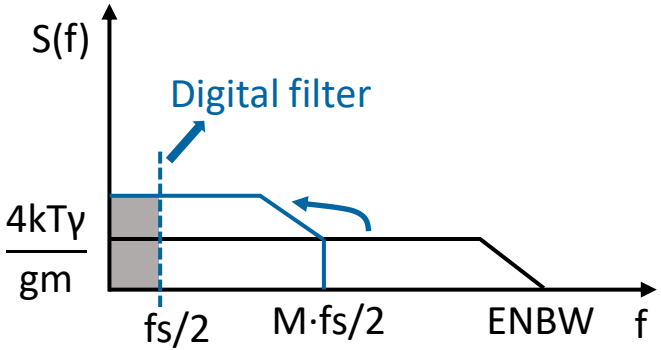




Proposed CT 2nd-Stage with LSB repeating

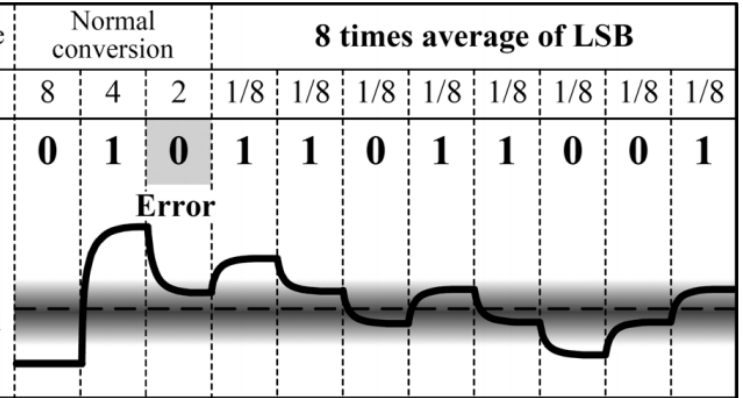
- Aliased noise in D_T can be averaged by digital filter





Prior LSB Repeating Technique: Tracking averaging

SAR state	Normal conversion			8 times average of LSB							
Weight	8	4	2	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8
C_{OUT}	0	1	0	1	1	0	1	1	0	0	1
D_{DEC}	-	-	-	1	2	2	2	3	3	2	2
Decoded value	0	4	0	D_{AVE_T} = 2.125							



[JSSC 2015, T. Miki]

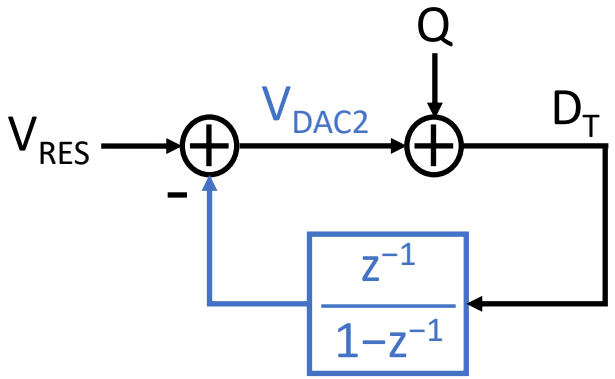
- Minimal hardware overhead
- PVT-robust noise suppression
- Complex reconstruction logic
- Behavioral analysis only

$$D_{DEC}[k] = \begin{cases} D_{DEC}[k-1] + C_{OUT}[k-1] + C_{OUT}[k] - 1, & k > 1 \\ C_{OUT}[k], & k = 1. \end{cases}$$

$$D_{AVE_T} = \sum_{k=1}^{M} \frac{D_{DEC}[k]}{M}$$

Proposed Optimized Tracking Averaging

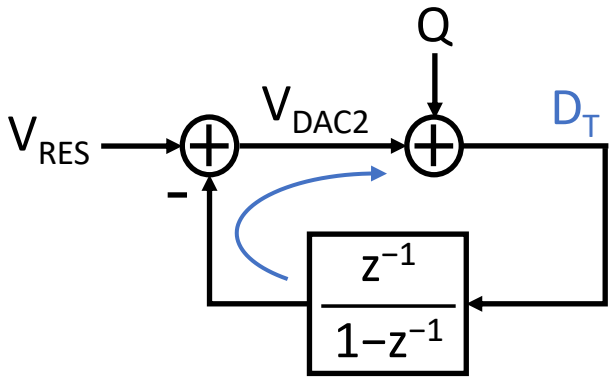
w/o thermal noise:



- Accumulated D_T is fed back to the V_{DAC2}

Proposed Optimized Tracking Averaging

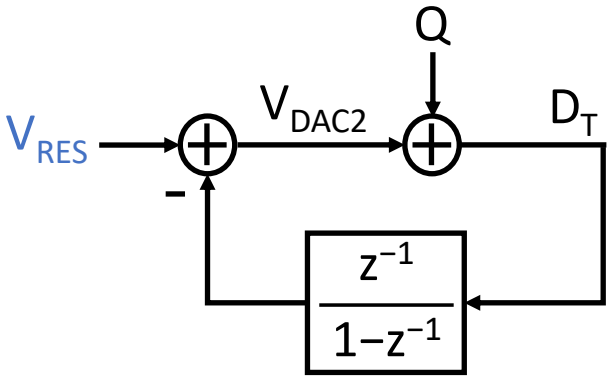
w/o thermal noise:



- Each D_T depends on V_{RES} and previous accumulated D_T

Proposed Optimized Tracking Averaging

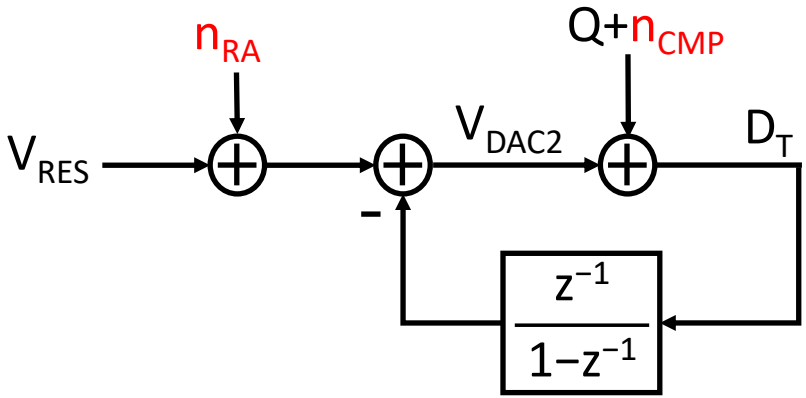
w/o thermal noise:



- 2nd-order accumulation of D_T reconstructs V_{RES}:
- $\sum_{k=1}^{M}\sum_{i=1}^{k} D_T(i) = M \cdot V_{RES} + \sum_{i}^{M} Q(i)$

Proposed Optimized Tracking Averaging

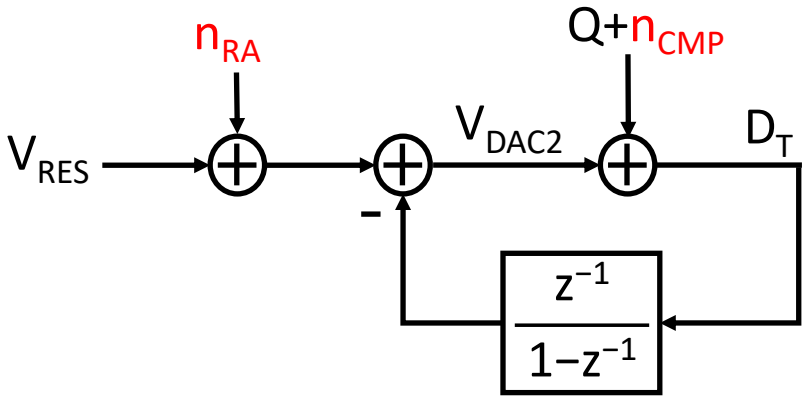
w/ thermal noise:

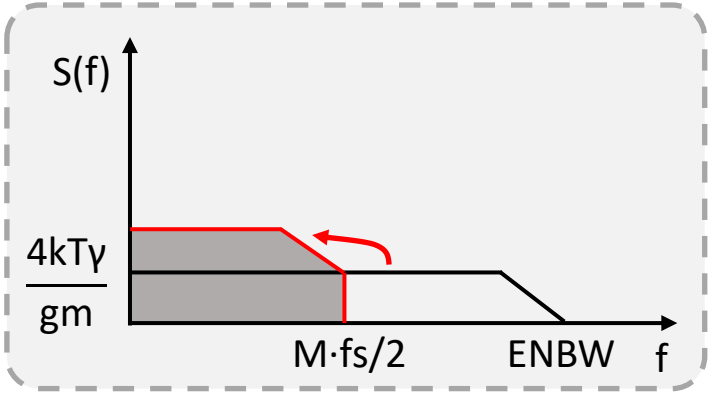


- Noise terms enter the same accumulation path:
- $\sum_{k=1}^{M}\sum_{i=1}^{k} D_T(i) = M \cdot V_{RES} + \sum_{i}^{M}[n_{RA}(i) + n_{CMP}(i) + Q(i)]$

Proposed Optimized Tracking Averaging

w/ thermal noise:

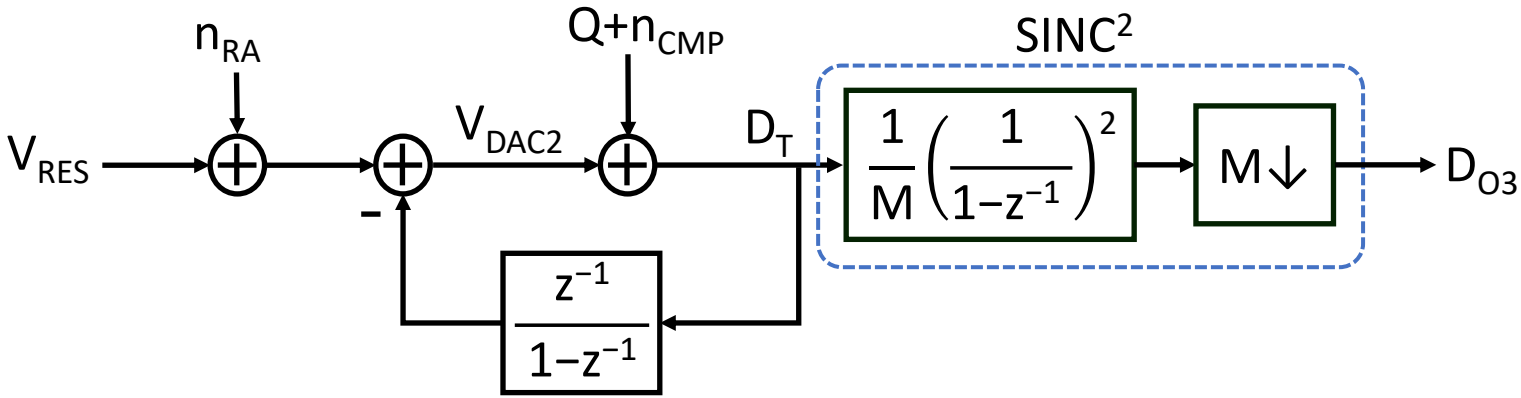




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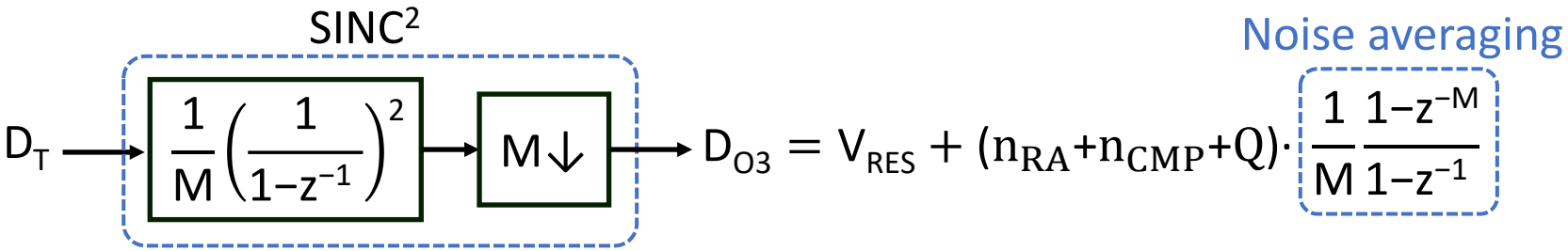
Proposed Optimized Tracking Averaging

w/ thermal noise:



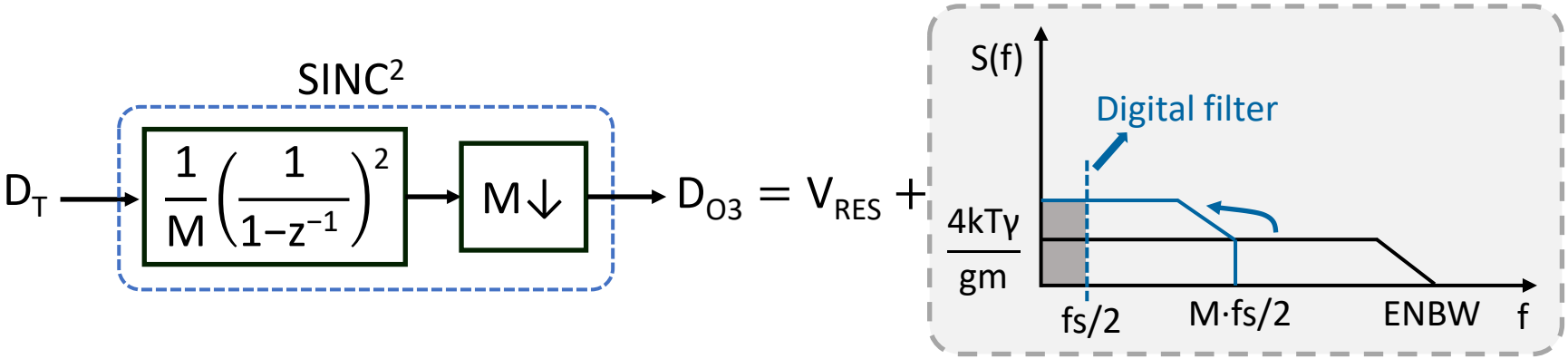
- Averaging and decimation give the final output:
- $D_{O3} = \frac{1}{M}\sum_{k=1}^{M}\sum_{i=1}^{k} D_T(i)$

Proposed Optimized Tracking Averaging



- Noise terms are averaged by M times
- $D_{O3} = V_{RES} + \frac{1}{M}\sum_i^M [n_{RA}(i) + n_{CMP}(i) + Q(i)]$

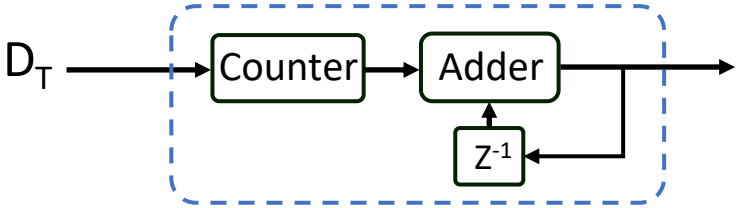
Proposed Optimized Tracking Averaging



- Noise terms are averaged by M times
- $D_{O3} = V_{RES} + \frac{1}{M}\sum_i^M [n_{RA}(i) + n_{CMP}(i) + Q(i)]$

Proposed Optimized Tracking Averaging

2nd–order CoI filter

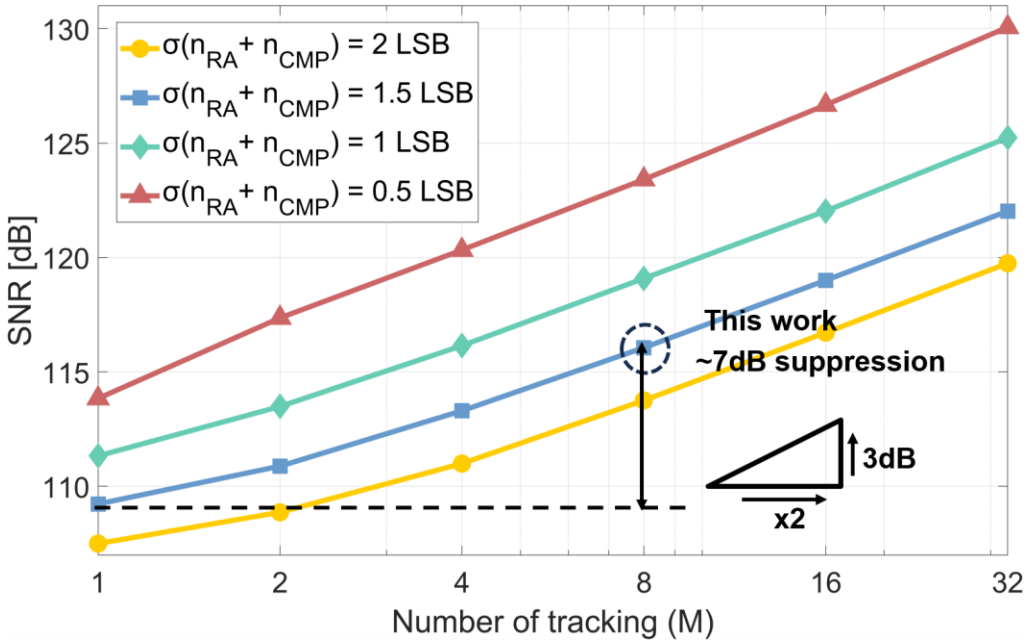


$$D_{O3} = V_{RES} + (n_{RA}+n_{CMP}+Q)\cdot\frac{1}{M}\frac{1-z^{-M}}{1-z^{-1}}$$

Noise averaging

- Simple 2nd-order CoI filter ☺
- Effective noise averaging ☺

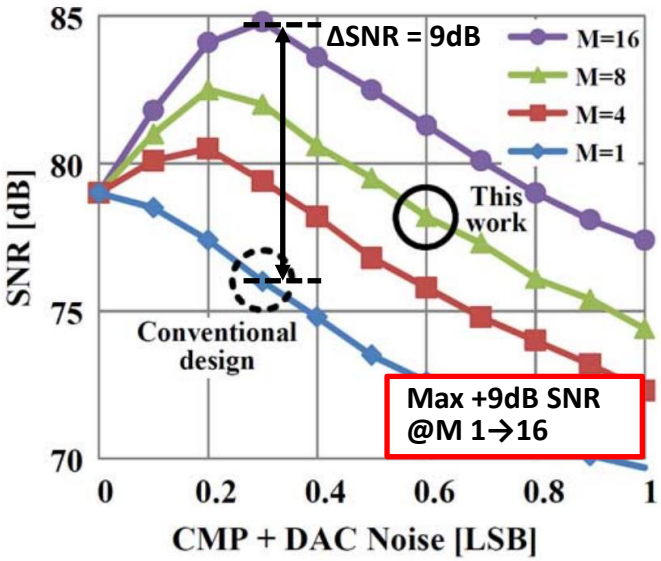
Simulation Results



- 18-bit ADC simulation including RA and comparator noise
- 3-dB SNR boost per doubling of M
- Robust noise reduction

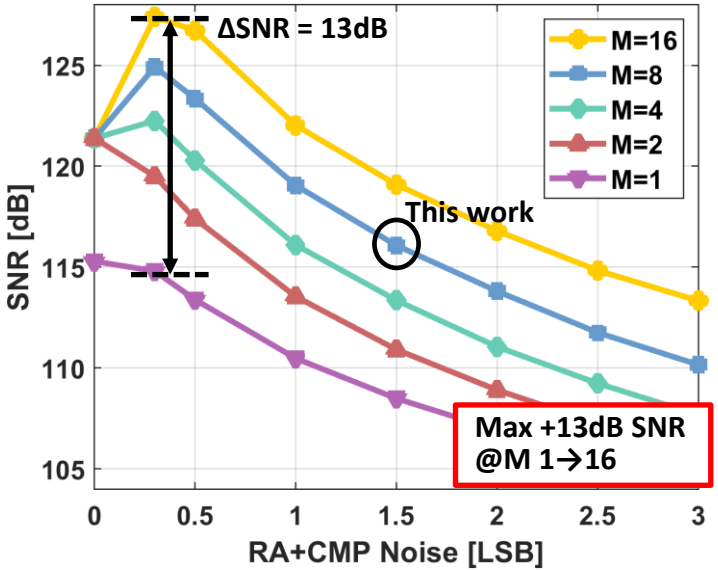
Comparison with Prior Work

Tracking Averaging: ΔSNR_{MAX}=9dB



[JSSC 2015, T. Miki]

Optimized: ΔSNR_{MAX}=13dB

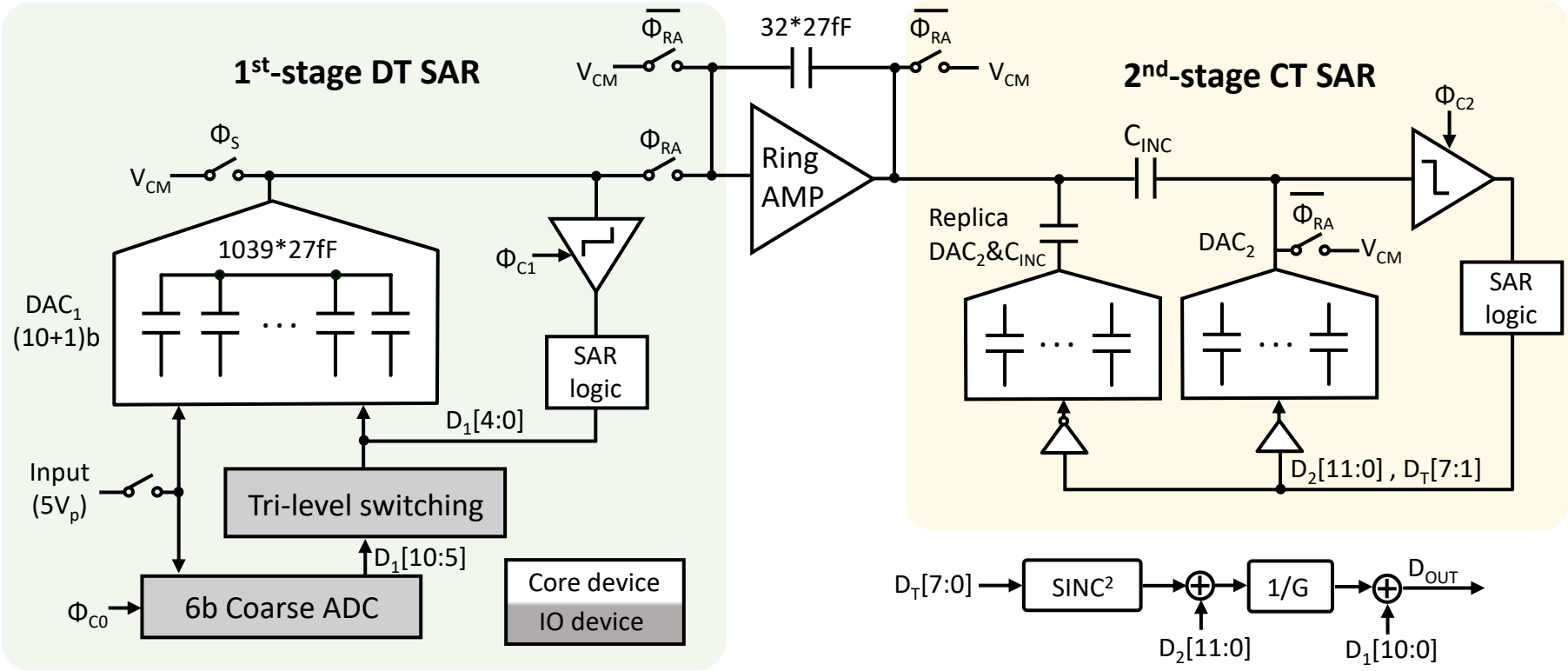


- Superior SNR boost under severe noise

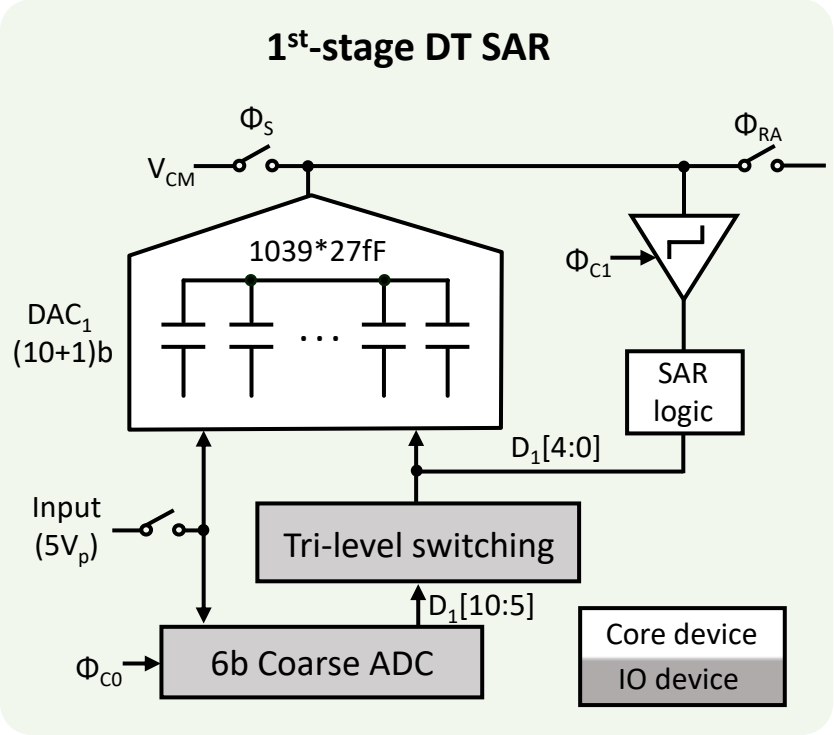
Outline

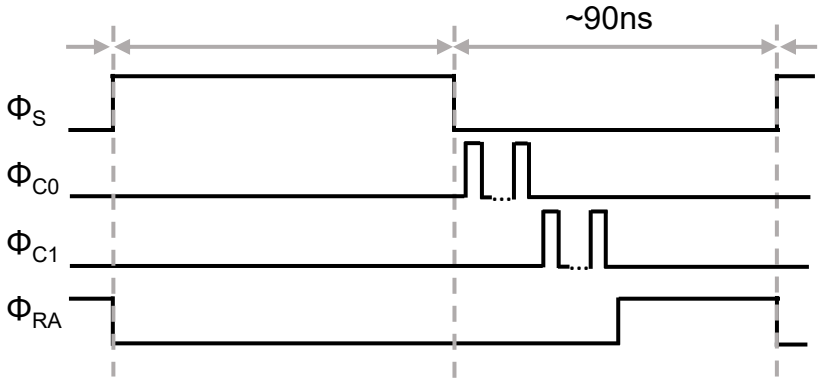
- Motivation
- Proposed 2-Step SAR with CT 2nd-Stage
- **Circuit Implementation**
- Measurement Results
- Conclusion

2-Step SAR ADC Implementation



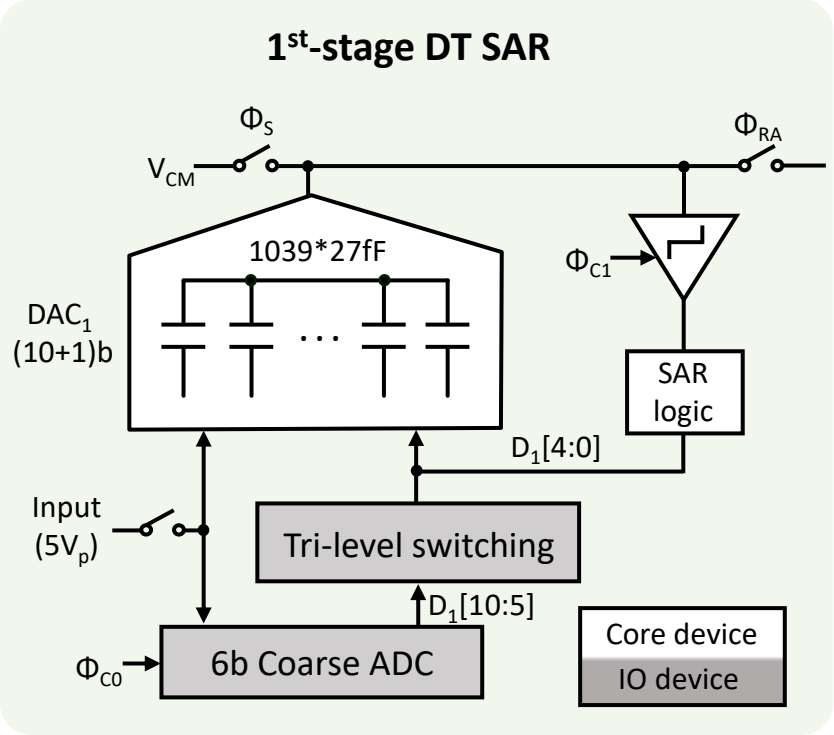
1st–Stage DT SAR

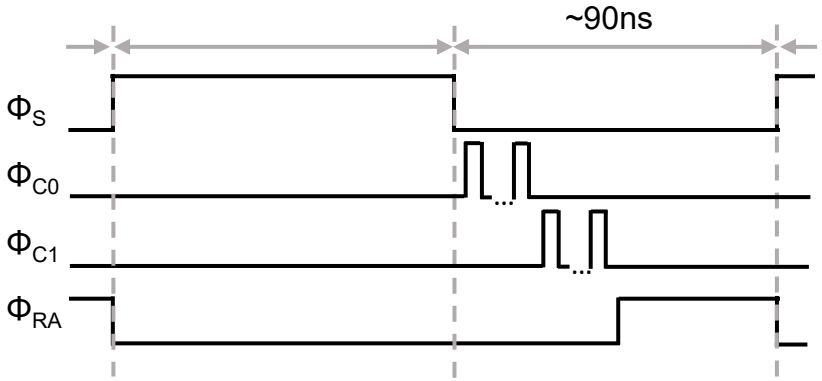




- 6-bit coarse ADC pre-resolves MSBs
- Tri-level switching in MSBs to relax I_{REF} burden

1st–Stage DT SAR

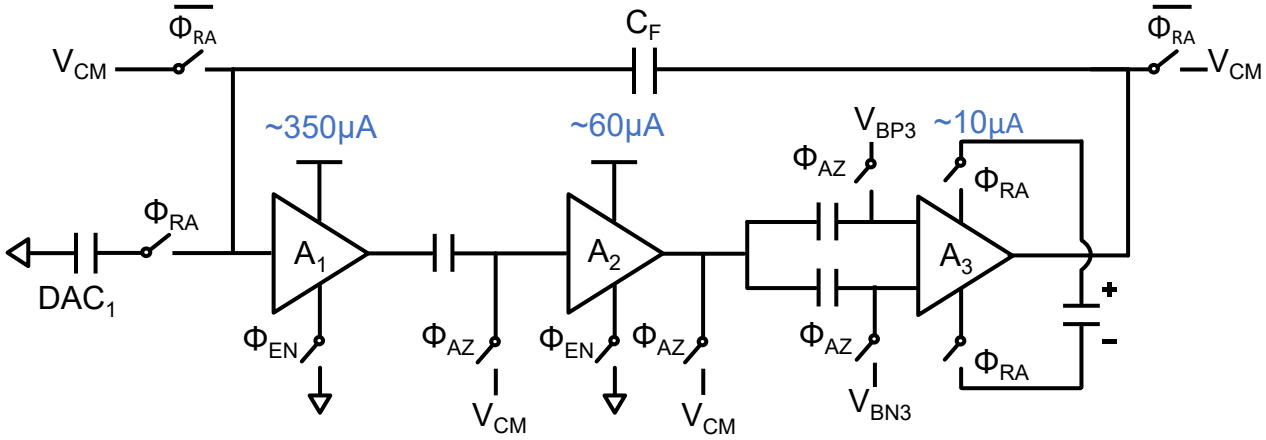


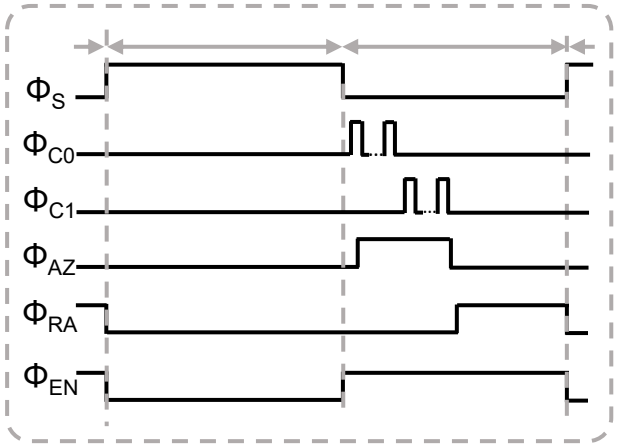


- ADC conversion time is fixed ~90ns
- Over 100ns sampling time:
  - ✓ ensures >20-bit linearity
  - ✓ relaxes input driving burden

Ring AMP Implementation

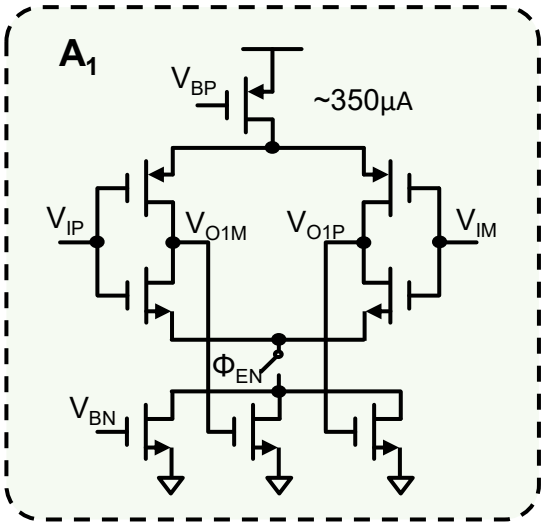
- Power off during ADC sampling
- Auto-zeroing concurrent with 1st-stage

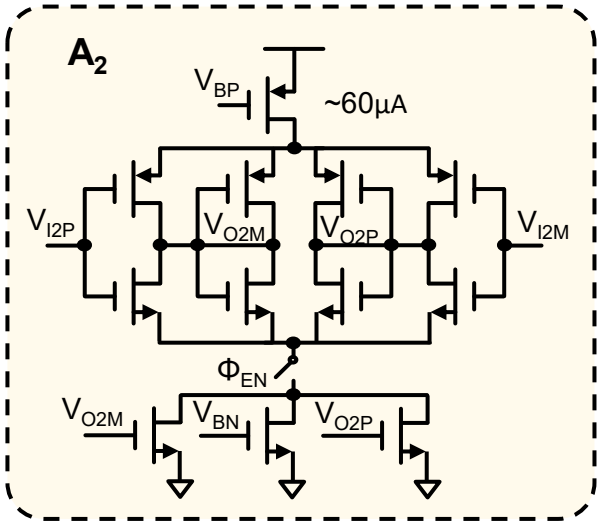




Ring AMP Implementation

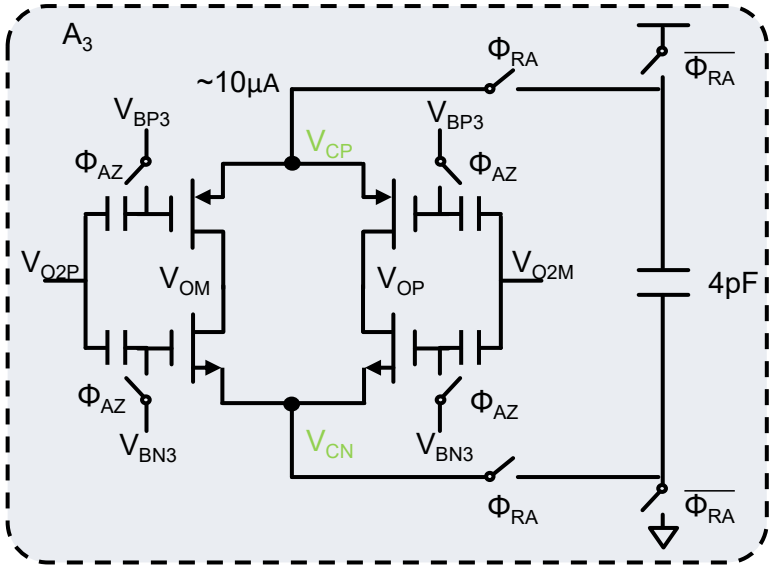
- Low current A_1 due to CT 2nd–stage noise reduction
- A_2 with diode-connected load for higher pole frequency and stability

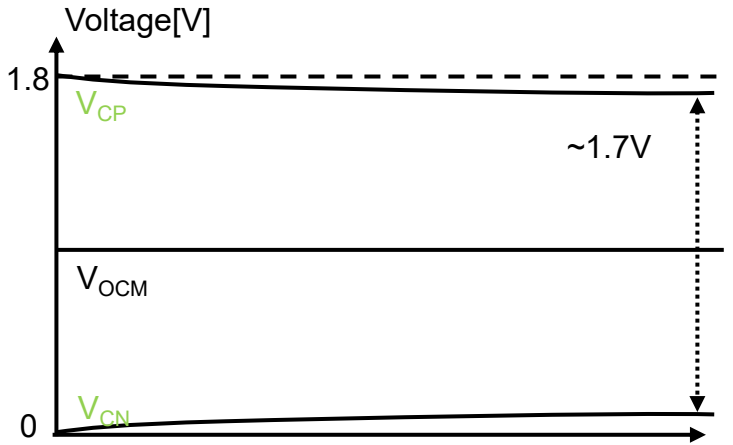




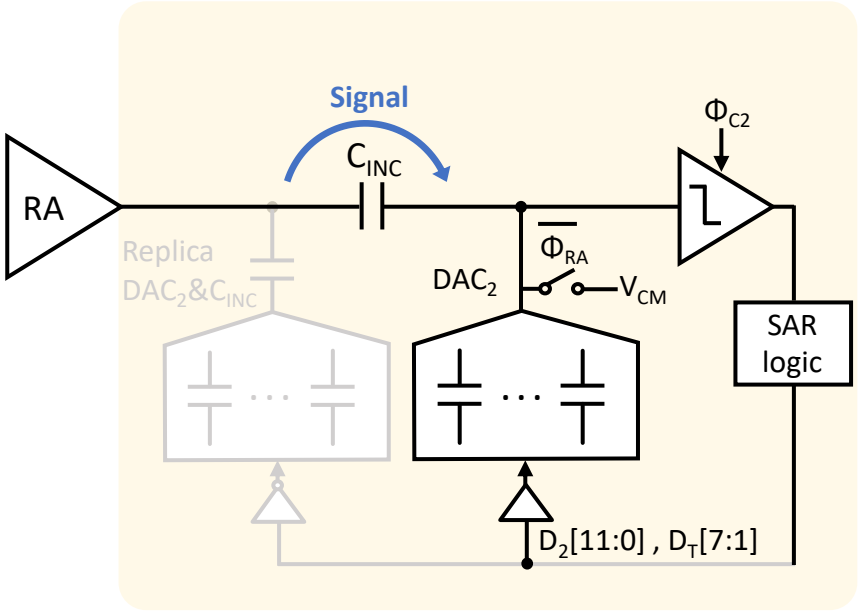
Ring AMP Implementation

- Biased FIA as output stage
  - ✓No need of global CMFB
  - ✓Large output swing



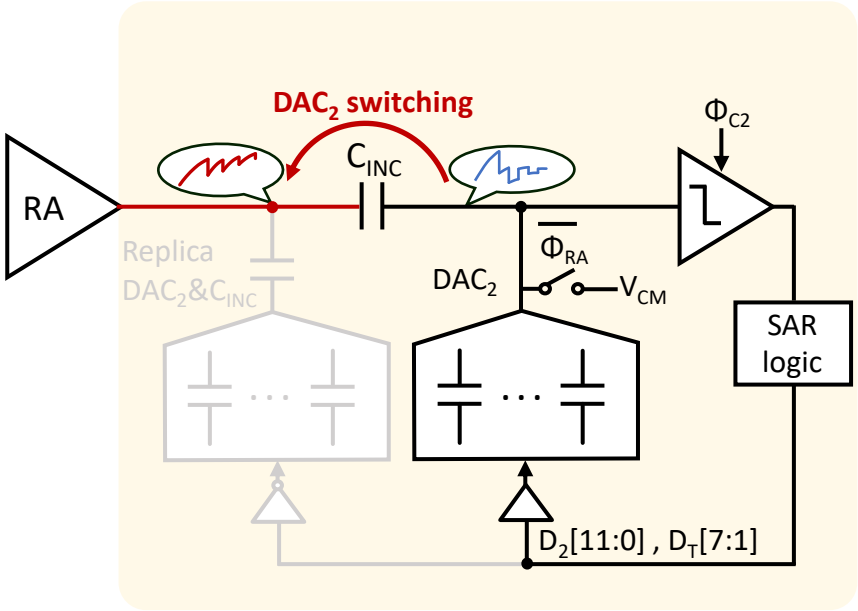


2nd–Stage CT SAR: DAC_2 Kickback

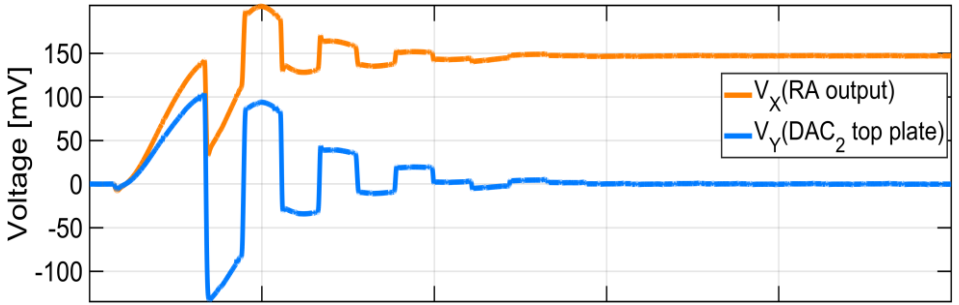


- RA output couples to DAC_2 top-plate through C_{INC}
- RA has limited output impedance

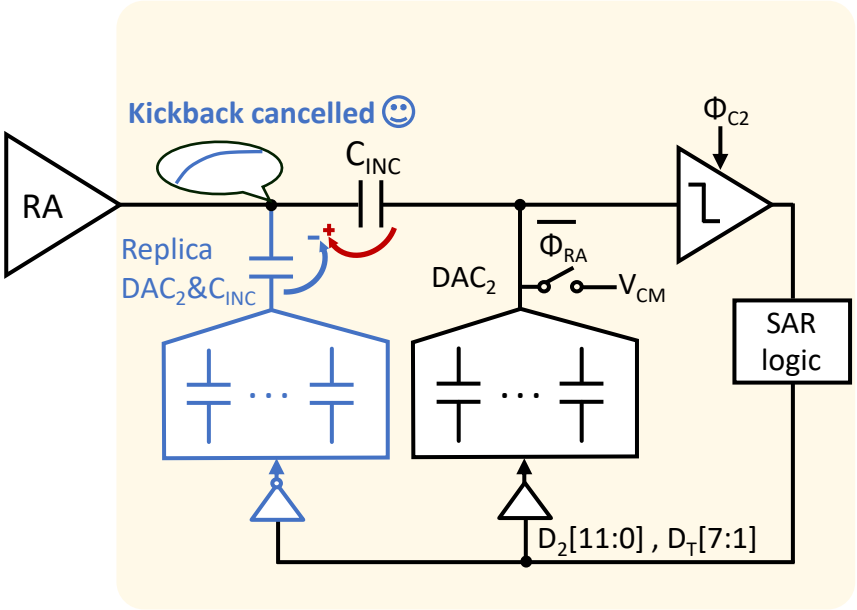
2nd–Stage CT SAR: DAC_2 Kickback



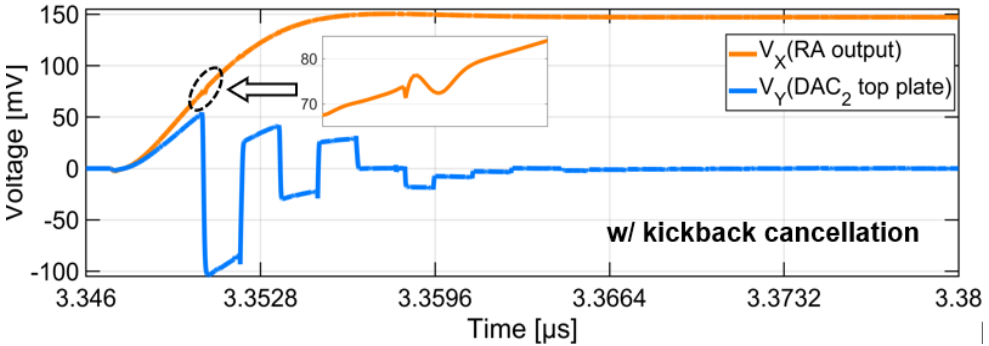
- DAC_2 kickback effects RA output
- Leading to wrong conversion and affect settling



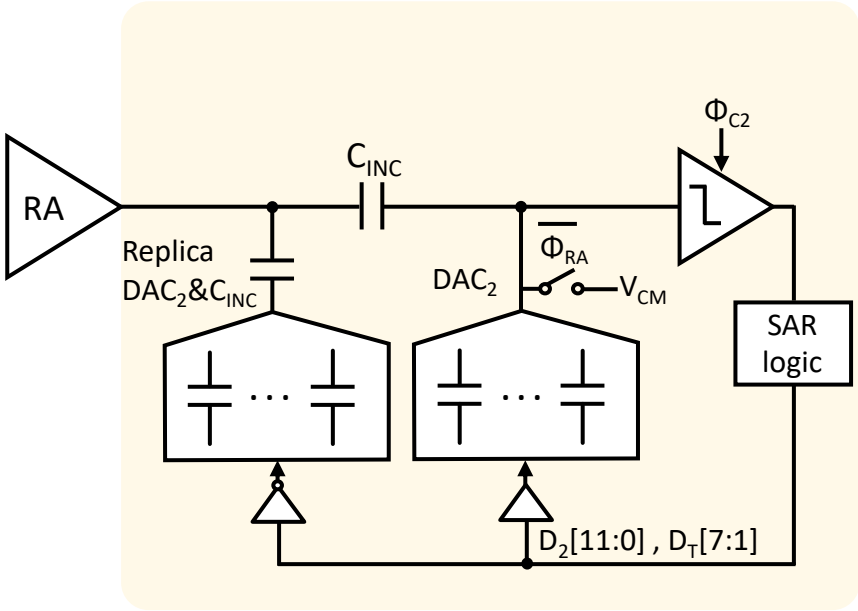
2nd–Stage CT SAR: Kickback Cancellation



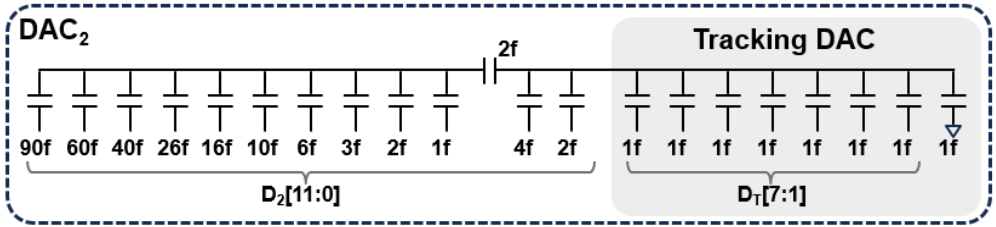
- Adding replica DAC_2&C_{INC} at RA output
- Timing carefully matched



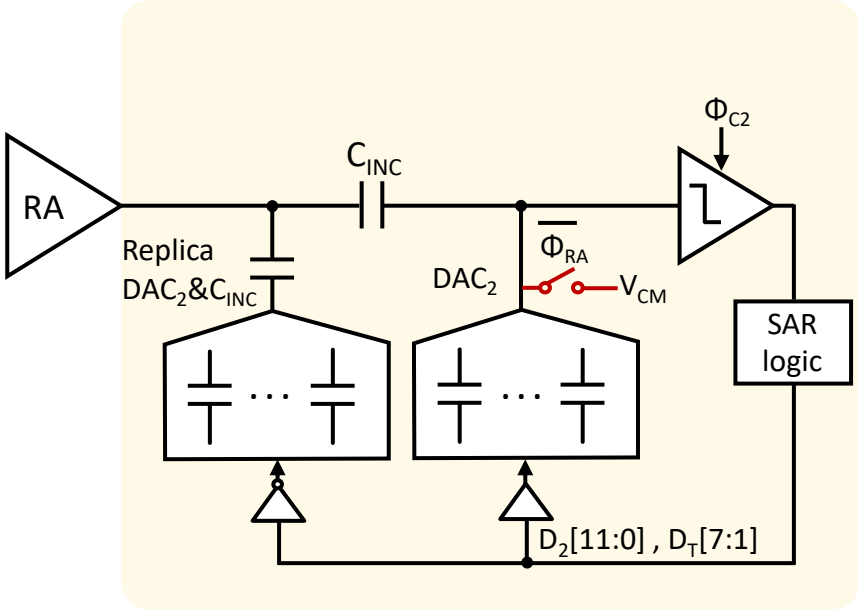
2nd–Stage CT SAR



- Massive redundancy in DAC_2 to tolerate RA settling errors

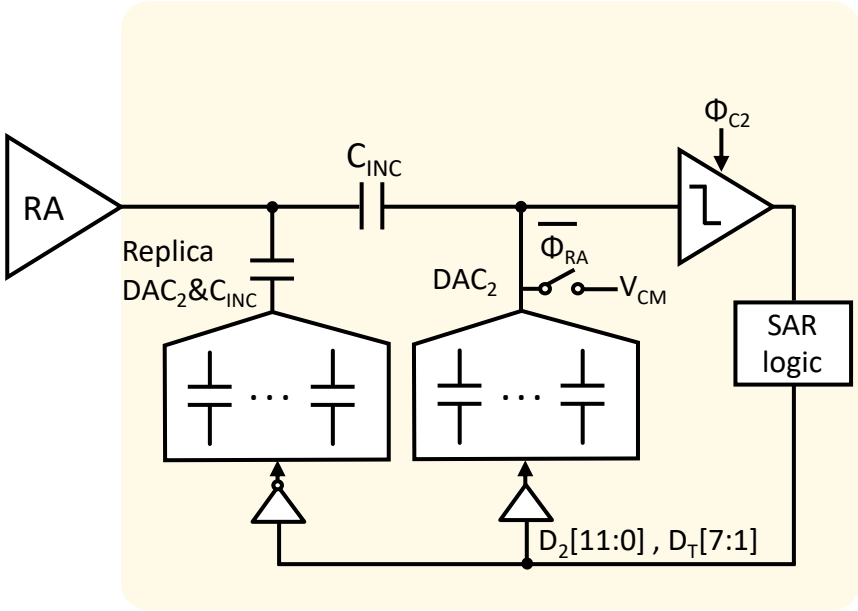


2nd–Stage CT SAR

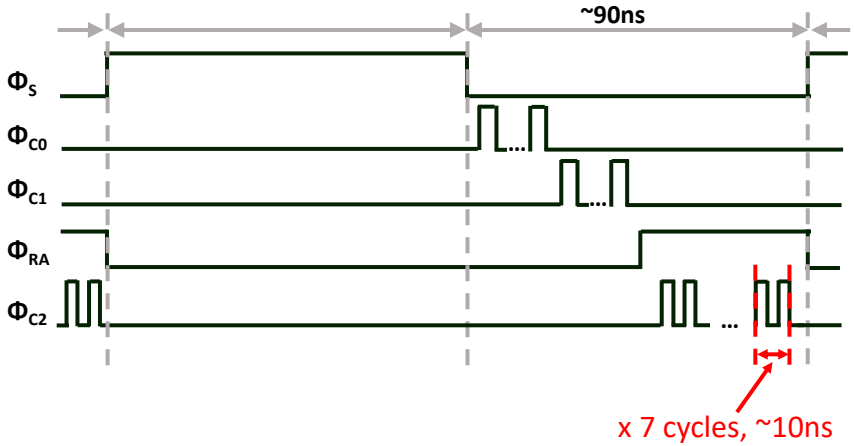


- Reset-switch kT/C noise is negligible due to RA gain

2nd–Stage CT SAR



- 8 LSB repeating cost an extra 10ns (~5% conversion time for +2dB SNR)

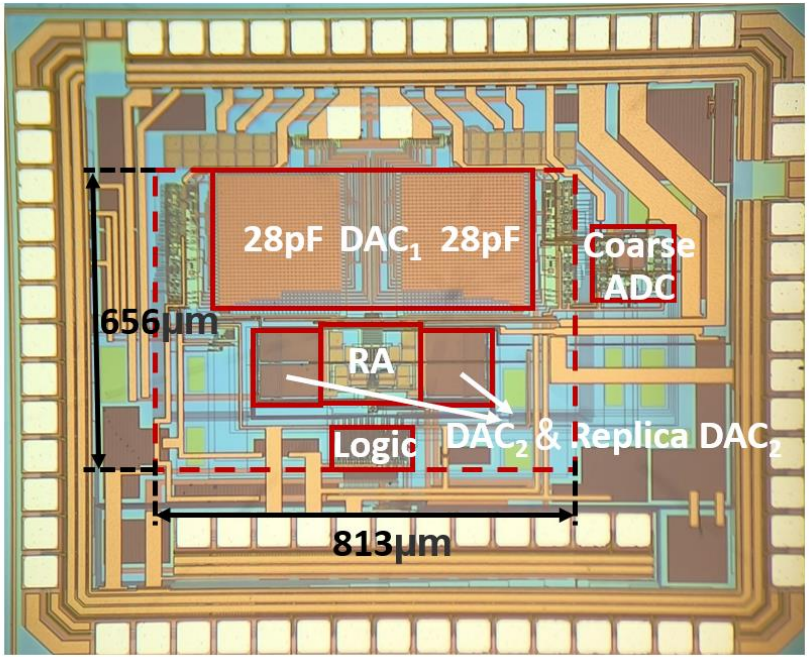


Outline

- Motivation
- Proposed 2-Step SAR with CT 2nd-Stage
- Circuit Implementation
- **Measurement Results**
- Conclusion

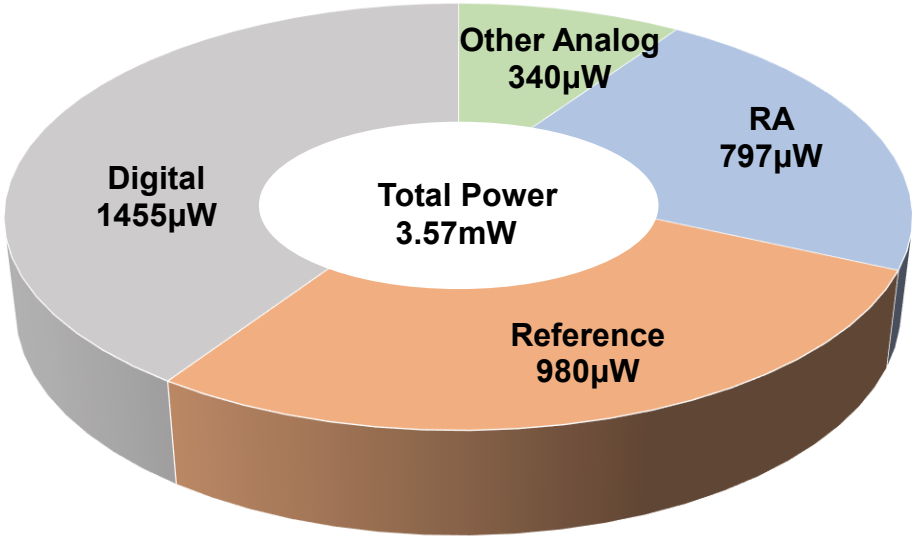
Chip Photograph

- **Process:** 180-nm BCD
- **ADC core area:** 0.53mm^2



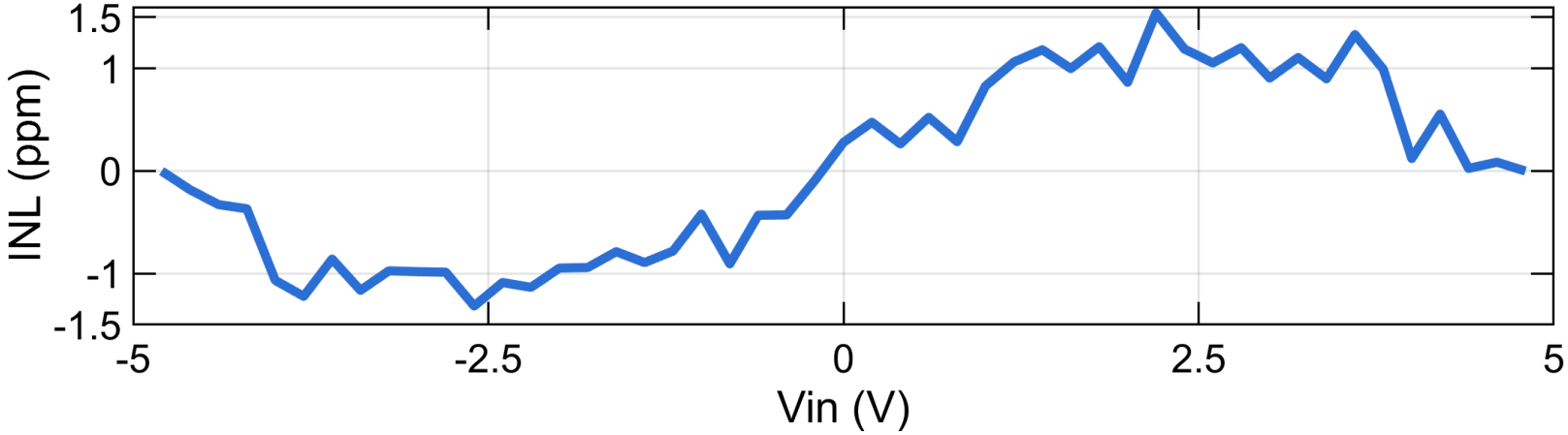
Power Breakdown

- **Total power** = 3.57mW



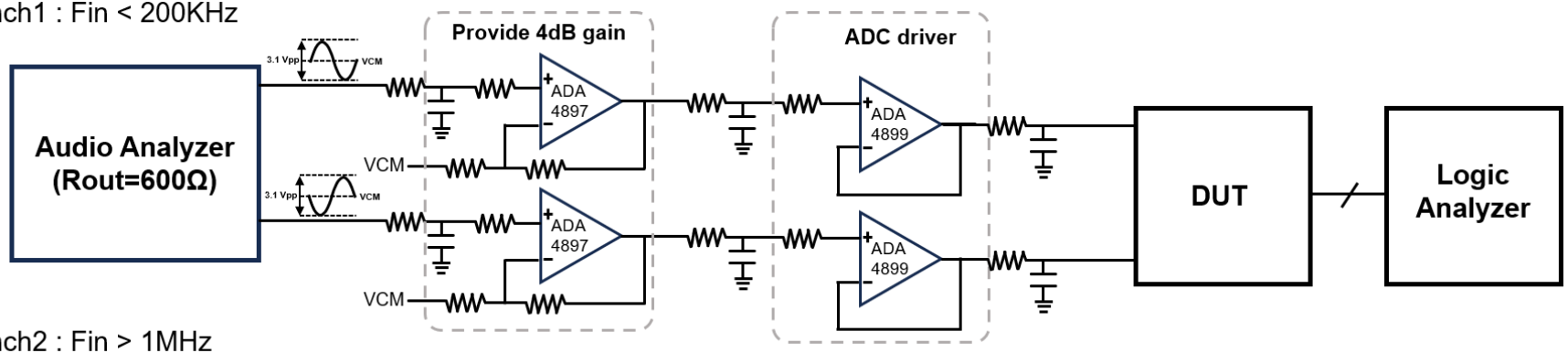
Measured INL

- INL=±1.5ppm

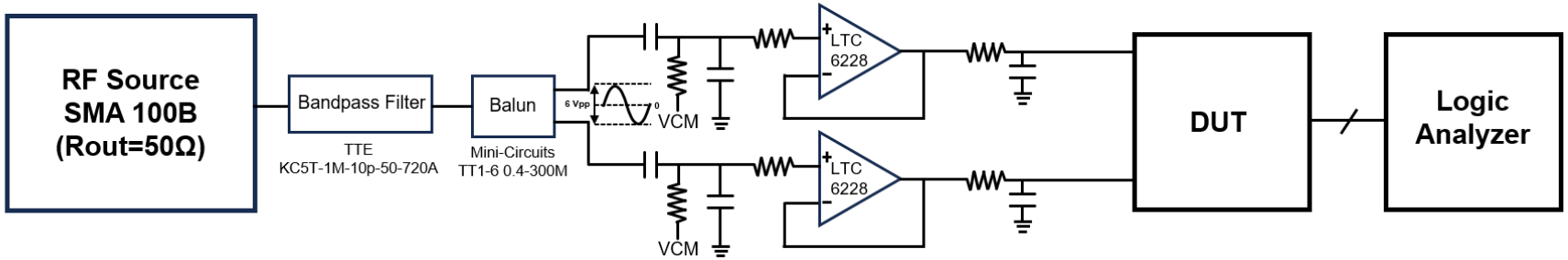


AC Measurement Setup

Bench1 : Fin < 200KHz

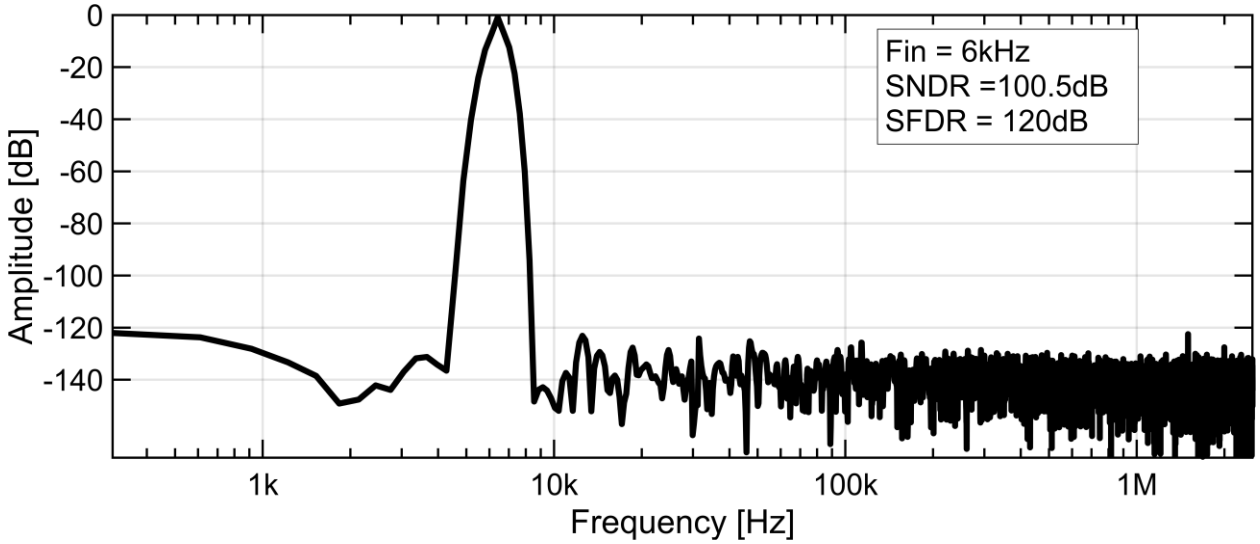


Bench2 : Fin > 1MHz



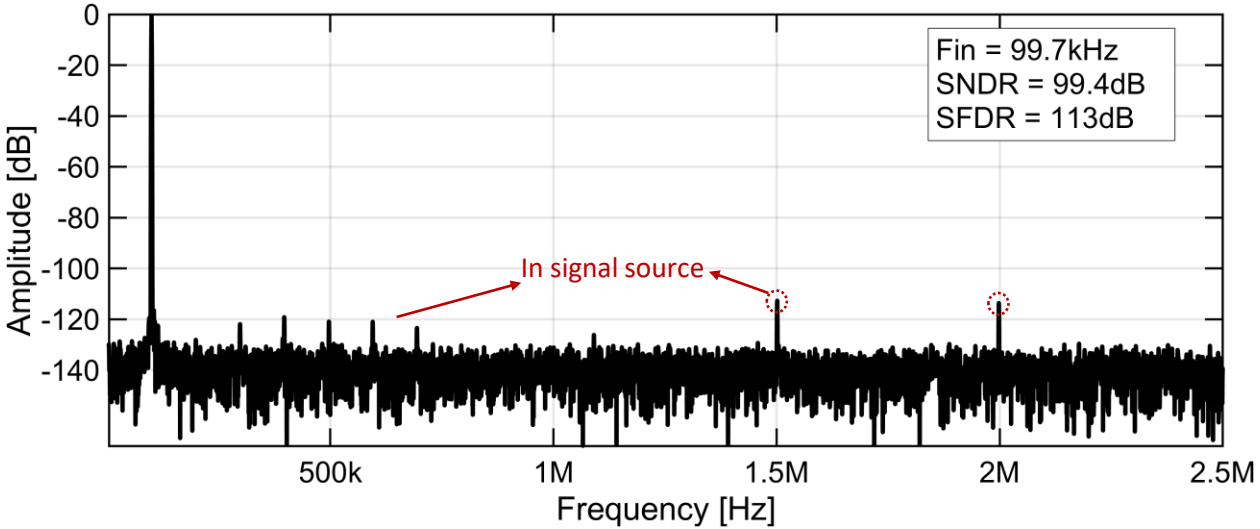
Measured Spectrum

- With one-time LMS mismatch calibration
- Fin = 6kHz



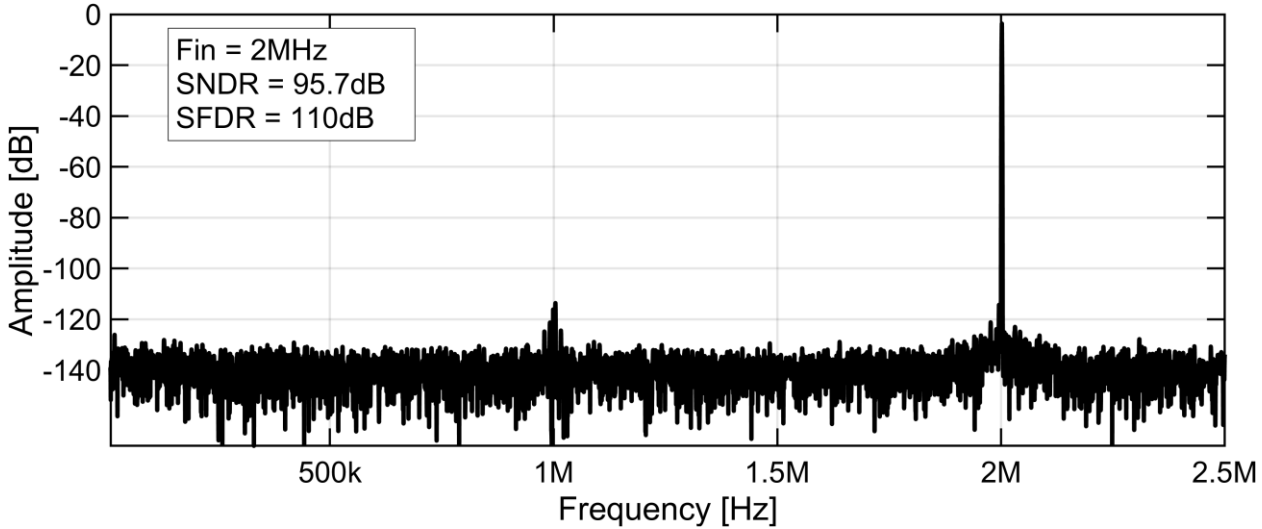
Measured Spectrum

- With one-time LMS mismatch calibration
- Fin = 99.7kHz



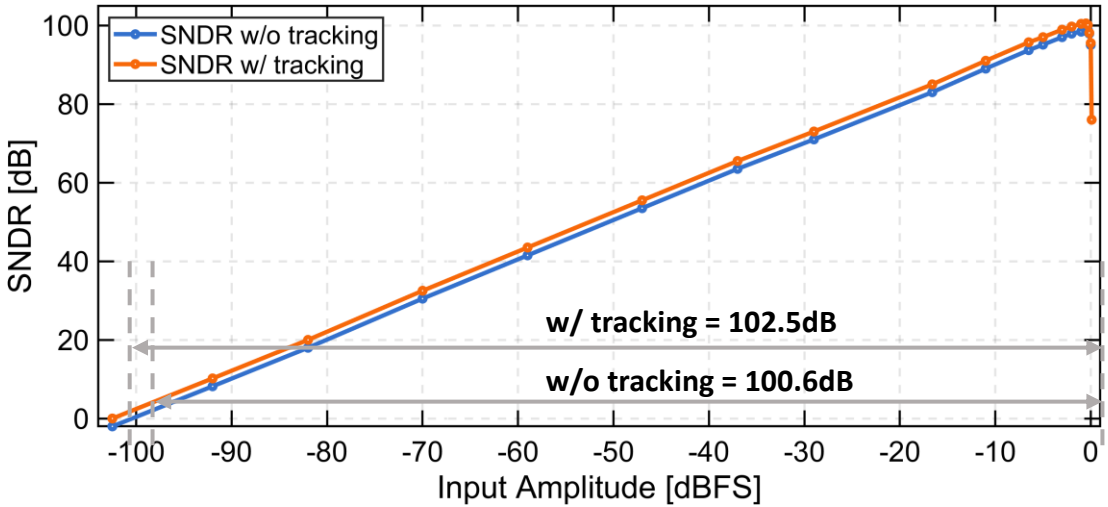
Measured Spectrum

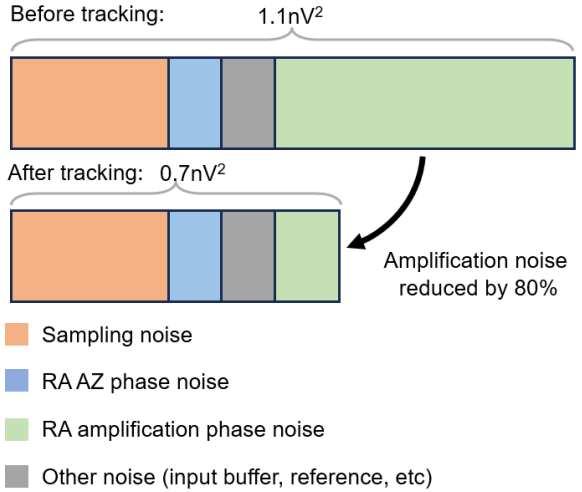
- With one-time LMS mismatch calibration
- Fin = 2MHz



Measured Dynamic Range

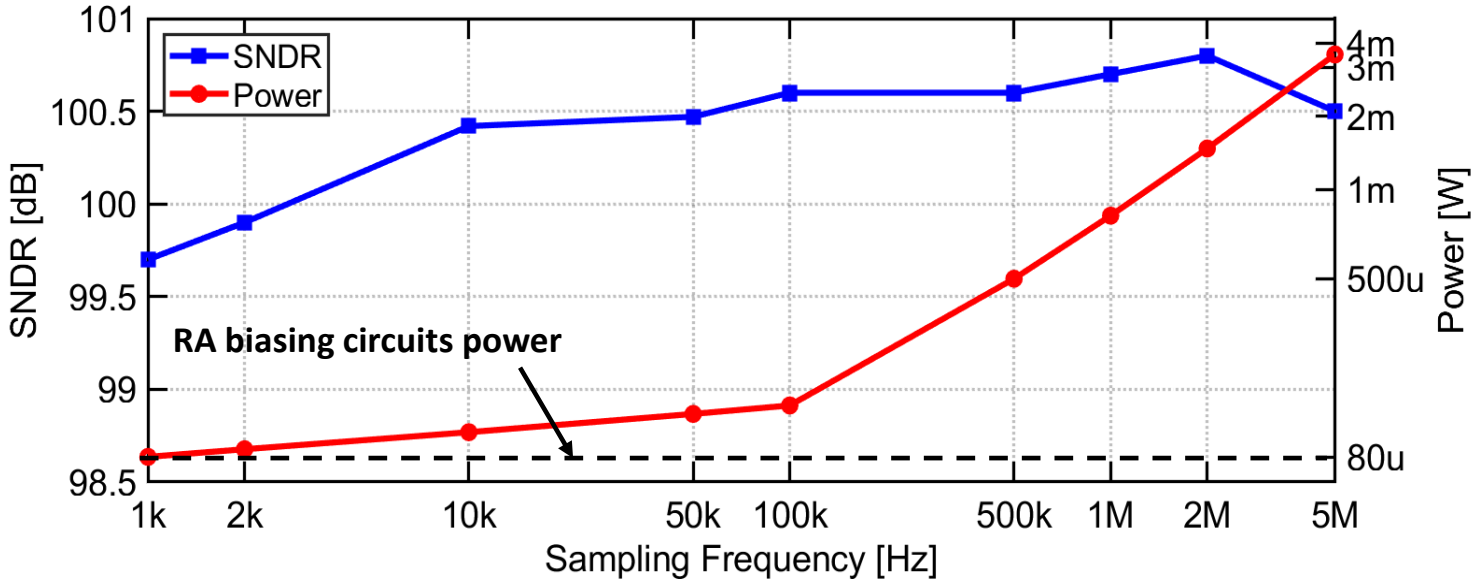
- DR = 102.5dB
- 8 times tracking average brings 80% RA noise reduction (by calculate)





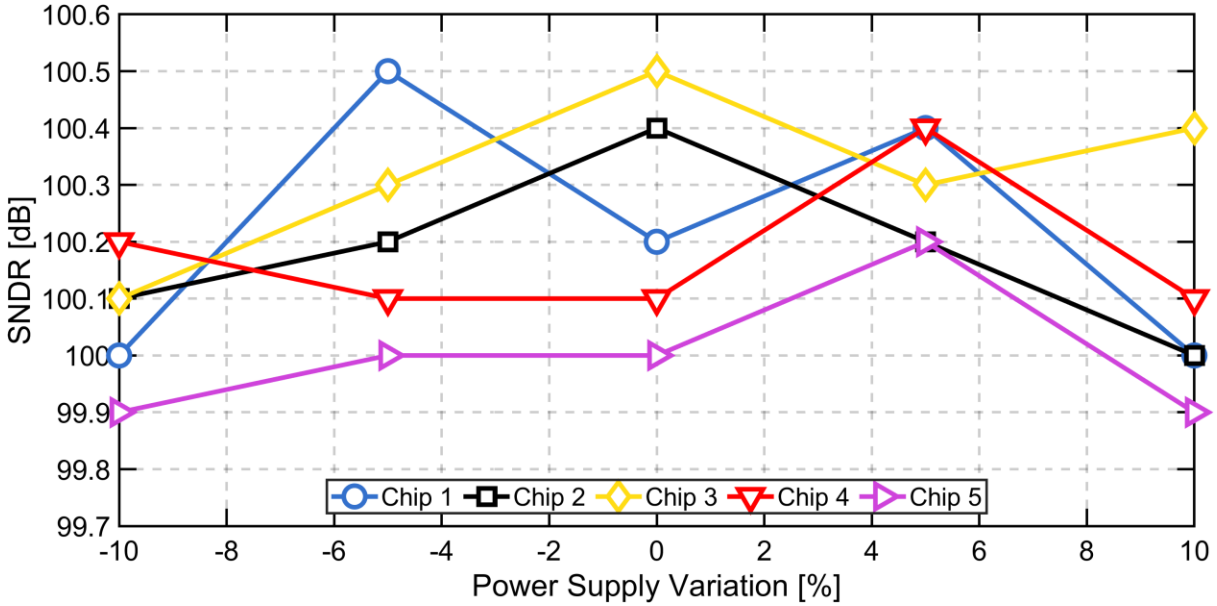
Measured SNDR & Power vs. Sampling Frequency

- Power consumption scales linearly vs. sampling rate
- Limited by RA biasing circuit power



Measured SNDR Across Chips & Supply

- ΔSNDR = 0.6dB



Performance Comparison

	This work	ISSCC 22 Steensgard	ISSCC 24 Bodnar	ISSCC 24 Ye	VLSI 14 Bannon
Architecture	**2-Step SAR**	Pipe SAR	TI-Pipe SAR	Pipe SAR	Pipe SAR
Noise/power reduction technique?	**CT 2nd-stage with tracking averaging**	Incomplete settling RA	2-phase settling RA	Bandwidth adaptive RA	-
RA noise reduction	**80%**	~75%	~40%	~50%	-
Process [nm]	**180**	180	40	22	180
Fs [MS/s]	**5**	2	40	200	5
Resolution [bit]	**20.5**	24	20	14	18
Area [mm²]	**0.53**	0.78	0.9	0.019	5.74
Supply [V]	**5 /1.8**	5 /1.8	3.3 /1.1	0.9	5 /1.8
C_S [pF]	**28**	50	20.5	1	-
Power [mW]	**3.6**	8.5	50	2	30.5
DR [dB]	**102.5**	106.3	94.2	72.8	100.2
SNDR[dB] @Fin [Hz]	**99.4@100k 95.7@2M**	105.3@100k	93.7@1k 92@1M	70.9@22M 70.7@94M	98.6@1k
FoM_S [dB] @Fin [Hz]	**187.8@100k 184.1@2M**	186.0@100k	179.7@1k 178@1M	177.9@22M 177.7@94M	177.7@1k

Performance Comparison

	This work	JSSC 25 Huang	ISSCC 19 ElShater	AD4020	LT2378
Architecture	**2-Step SAR**	SAR	Pipe SAR	SAR	SAR
Noise/power reduction technique?	**CT 2nd-stage with tracking averaging**	SRM	-	-	-
RA noise reduction	**80%**	-	-	-	-
Process [nm]	**180**	180	180	-	-
Fs [MS/s]	**5**	5	15	1.8	1
Resolution [bit]	**20.5**	16	16	18	20
Area [mm^2]	**0.53**	0.57	1.82	-	-
Supply [V]	**5 /1.8**	3.3 /1.8	3.3 /1.8	5 /1.8	5 /2.5
C_S [pF]	**28**	21	17	40	45
Power [mW]	**3.6**	5.3	10	15	21
DR [dB]	**102.5**	95	95	101	-
SNDR[dB] @Fin [Hz]	**99.4@100k 95.7@2M**	93.7@2k	91@50k 88@4M	96.5@100k	102@100k
FoM$_S$ [dB] @Fin [Hz]	**187.8@100k 184.1@2M**	180.4@2k	179.8@50k 176.8@4M	174.3@1k	175.8@100k

Outline

- Motivation
- Proposed 2-Step SAR with CT 2nd-Stage
- Circuit Implementation
- Measurement Results
- **Conclusion**

Conclusion

- Proposed a high-precision 2-Step SAR ADC
  - A 20-bit 5-MS/s SAR ADC in 180-nm CMOS.
- Introduced a CT 2nd-Stage with Tracking Averaging
  - Breaks the traditional tradeoff between residue amplifier settling and noise bandwidth.
- Significant Noise Efficiency Improvement
  - 80% RA noise reduction by only 5% more time.
- State-of-the-Art Performance
  - Delivers 102.5dB DR and ~100dB peak SNDR (99.4dB @ 100kHz).
  - Achieves a Schreier FoM of 187.8 dB